Time-Series PolSAR and Multispectral Fusion for Enhanced Hypersaline Water Body Classification

Fan Zhang, *Senior Member, IEEE*, Fanle Meng, Fei Ma, *Member, IEEE*, Qiang Yin, *Senior Member, IEEE*, and Yongsheng Zhou, *Member, IEEE*

***Abstract*— Classification of hypersaline water bodies, for example, salt fields and salt lakes, presents unique challenges due to the similar spectral and scattering characteristics of various saline water bodies. To address this issue, we propose an innovative classification method tailored for such environments, integrating Sentinel-1 multitemporal polarimetric synthetic aperture radar (PolSAR) data with Landsat multispectral imagery. The method introduces a novel PolSAR-based feature, termed scattering mechanism entropy, to quantify variations in salt crystal precipitation processes. Additionally, the blue band from multispectral imagery is leveraged to represent ion concentrations in hypersaline water bodies. By deriving the statistical relationship between scattering mechanism entropy and blue band data for each pixel, we amplify the separability of salt features and mitigate the influence of spectral similarity. These derived features are then concatenated into a Chernoff distance-based classifier for improved classification performance. To validate the robustness and effectiveness of this method, we apply it to the classification of salt fields and salt lakes on four major salt lake sites in China: Qarhan salt lake (2018–2020), Yiliping salt lake (2021–2023), Taijnar salt lake (2019–2023), and Gasikule salt lake (2019–2023). The proposed approach achieves classification accuracies of 90.91%, 92.73%, 97.40%, and 97.65%, respectively, significantly outperforming existing water body classification methods.**

***Index Terms*— High-salinity environments, multispectral imagery, polarimetric synthetic aperture radar (PolSAR), scattering mechanism entropy, water body classification.**

I. Introduction

HYPERSALINE water bodies, characterized by salinity levels exceeding 3.5%, play critical roles in both ecological systems and mineral resource extraction [1]. Among these, salt lakes represent natural hypersaline water bodies rich in lithium, potash, magnesium, and other strategic minerals, while salt fields are artificial hypersaline facilities designed to concentrate and harvest these resources through controlled evaporation processes. Fig. 1 illustrates the production process of the salt mine. The brine from the salt lakes is initially pumped into sodium salt fields. As the brine evaporates and concentrates, sodium salt crystals start precipitating. Once these crystals form on the surface, the remaining brine gets transferred to the carnallite fields. In carnallite fields, continued evaporation precipitates carnallite ($KMgCl_3{\cdot}6H_2O$), a valuable mineral rich in potassium and magnesium, at the pool bottoms. The carnallite is then transported to the factory for processing, while the remaining brine is returned to the salt lakes after additional processing. Consequently, the salt lake regions consist of two main land-cover types: salt lakes and salt fields, while salt fields can be subdivided into carnallite fields and sodium salt fields. This dynamic interplay between natural salt lakes and engineered salt fields defines hypersaline water body ecosystems.

Accurate classification of hypersaline water bodies facilitates monitoring of area changes, crucial for ecological protection and resource management. For example, analysis of salt lake area fluctuations serves as an indicator of ecosystem health, reflecting changes in water levels, salinity, and habitat integrity. This information can guide the implementation of targeted conservation measures. Concurrently, tracking salt field area changes enables assessment of mineral resource availability and sustainability, informing resource management strategies [2]. These data provide government authorities with a quantitative basis for balancing resource utilization and environmental preservation.

The classification of hypersaline water bodies faces unique challenges due to their overlapping spectral and scattering properties. The traditional classification of salt lake regions primarily relies on manual field research, which requires a great deal of human and material resources [3]. The rapid advancement of remote sensing technology offers a more efficient alternative, which has been widely utilized in land cover classification due to its extensive coverage, timely observations, and cost-effectiveness [4]. Currently, the main data sources for remote sensing are optical images and synthetic aperture radar (SAR) images. Optical images, with their rich spectral features and texture information, enable precise classification of various land covers. For example, Long et al. [5] used convolutional neural networks to extract deep features from optical images for crop classification, leveraging both spectral and textural information. Similarly, Zhao et al. [6] employed a Pyramid Scene Parsing Network to capture contextual information from various regions within optical images, enhancing classification accuracy. Despite their advantages,

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Fan Zhang is with the College of Information Science and Technology and the Interdisciplinary Research Center for Artificial Intelligence, Beijing University of Chemical Technology, Beijing 100029, China (e-mail: zhangf@mail.buct.edu.cn).

Fanle Meng, Fei Ma, Qiang Yin, and Yongsheng Zhou are with the College of Information Science and Technology, Beijing University of Chemical Technology, Beijing 100029, China (e-mail: 2022210544@buct.edu.cn; mafei@mail.buct.edu.cn; yinq@mail.buct.edu.cn; zhyosh@buct.edu.cn).

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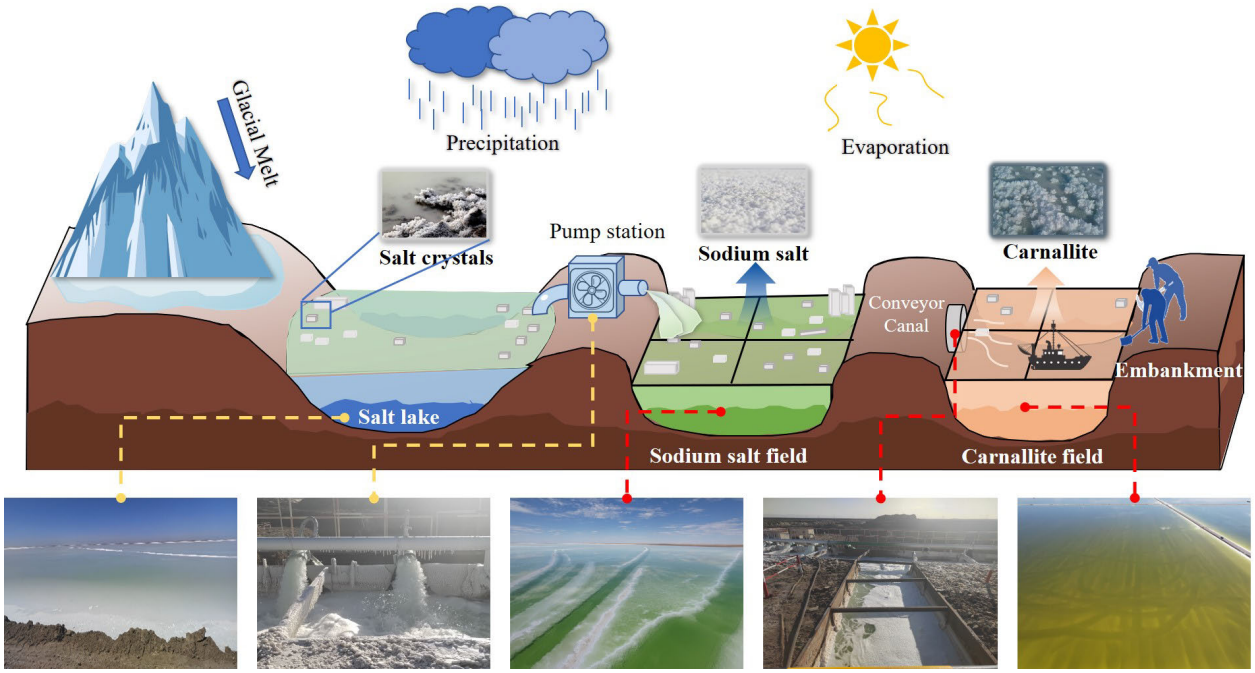


Fig. 1. Typical hypersaline water bodies: salt lakes, salt fields, and their subtypes.

optical images have limitations. They are highly sensitive to environmental conditions, such as weather and lighting, which can lead to inconsistent image quality. Therefore, optical images usually contain a large number of shadows, clouds, and noise, which seriously affect the accuracy of land cover classification.

SAR provides all-day, all-weather earth observation capabilities, overcoming optical satellites' environmental limitations [4]. Among all SAR sensors, polarimetric SAR (PolSAR) enhances traditional SAR by incorporating polarization information, revealing distinct scattering mechanisms of land covers. Exploiting these properties, Ma et al. [7] developed a superpixelwise building extraction method, while Dabboor and Shokr [8] derived the Wishart Chernoff distance for sea ice classification. However, these methods face challenges in salt lake regions due to the similar scattering characteristics of salt lakes and salt fields, both being water bodies with salt crystals attached to their surfaces. Consequently, accurate categorization of salt lake regions using conventional PolSAR land-cover classification methods remains problematic.

Field surveys indicate that different salt fields and lakes typically exhibit various salt ion concentrations, causing differential salt crystal accumulation rates. This leads to distinct temporal variations in land-cover characteristics between salt lakes and salt fields. These observed differences in change dynamics offer a novel approach for salt lake region classification. Multitemporal PolSAR data effectively capture dynamic land-cover changes, demonstrating superior classification performance, particularly for periodically changing landscapes [9]. Based on multitemporal PolSAR data, Ni et al. [10] proposed a multitemporal polarization feature, termed temporal eigenvalue, for land-cover classification in low-coherence scenarios. Experimental results demonstrated that temporal eigenvalues effectively enhance classification accuracy. Similarly, Gao et al. [11] extracted a kind of time-variant scattering feature from multitemporal PolSAR data to indicate types and directions of land-cover changes, successfully applied to crop classification. These advances suggest the potential of multitemporal polarization features for differentiating salt lakes and salt fields. Our previous work [12] proposed scattering mechanism entropy, a multitemporal polarization feature utilizing the entire PolSAR time series to capture cumulative land cover changes, enabling precise classification of salt lake regions. However, this method requires data spanning a sufficiently long period, typically over 12 months, to cover the entire process from brine to carnallite. The increase in the amount of data results in longer data preprocessing and feature calculation times. Reducing the time series to one production cycle (4–6 months) to improve efficiency compromises classification accuracy. With limited data, the scattering mechanism entropy fails to differentiate sodium salt fields from salt lakes. This is because insufficient data (4–6 PolSAR images) cannot adequately cover all stages of salt field change, so the scattering mechanism entropy is very similar for salt lakes and salt fields.

Based on the above analysis, multitemporal polarization features alone, while computationally efficient, are insufficient to differentiate between salt lakes and salt fields. To address this limitation, we propose integrating optical features with multitemporal polarization data. Salt fields typically exhibit shallower water depths and higher salt ion concentrations compared to salt lakes, resulting in distinct spectral scattering and absorption characteristics [13]. Combining optical and multitemporal polarization features can simultaneously leverage information on land-cover change dynamics and ion concentrations. This integrated approach enables high-precision

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classification of salt lakes and salt fields while maintaining computational efficiency.

Compared to existing methods, the main contributions of the proposed method are as follows.

1) We propose a fusion feature by combining the blue band from multispectral images, which reflects ion concentrations in salt waters, with the scattering mechanism entropy derived from PolSAR data, called polarimetric-optical salt dynamics feature (POSD). This integration leverages the complementary strengths of optical and radar data, enhancing the discriminative capability of the method.
2) We establish a statistical relationship between scattering mechanism entropy and blue band data, modeling their joint distribution using a Gaussian mixture model. This statistical framework amplifies the differences between salt lakes and salt fields, effectively mitigating the influence of spectral similarity. Furthermore, we employ a Chernoff distance-based classifier for precise classification, which improves the accuracy and robustness of the mapping results.

The remainder of this article is organized as follows. Section II introduces the proposed method. Section III describes the study area and data. Section IV describes the experimental process and results. Section V concludes this article.

II. Method

The framework of the proposed method is shown in Fig. 2. Section II-A describes how the POSD feature is extracted to represent the cumulative changes and ion concentrations of the salt lake region. The second part mathematically derives the POSD's probability distribution and proposes the classification method based on the Chernoff distance of the POSD feature.

A. Feature Extraction: POSD

1) Multitemporal Polarization Feature Extraction: In terms of multitemporal polarization features, several multitemporal polarization features are available, such as temporal eigenvalues [10], time-variant scattering features [11], and we previously proposed scattering mechanism entropy [12]. Among them, the scattering mechanism entropy can use all the data in the time series to calculate the cumulative change of land covers, while other multitemporal polarization features can only calculate the degree of change at two-time points. Therefore, we employ the scattering mechanism entropy to obtain a more accurate degree of change in land cover.

Let $\ell = \{\ell_t\}_{1\le t\le M}$ represent a time series of M preprocessed PolSAR images sorted by acquisition date, with ℓ_t the image at time t. $\mathbf{C}_t^{(i,j)}$ denotes the polarization covariance matrix of the pixel at coordinate (i, j) within the entire PolSAR image ℓ_t. Then, the temporal correlation $R_{l,g}^{(i,j)}$ of pixel (i, j) between two PolSAR images at time l and g can be calculated as

$$R_{l,g}^{(i,j)} = 2\frac{\sqrt{\left|\lambda_l^{(i,j)}\right|\left|\lambda_g^{(i,j)}\right|}}{\left|\lambda_l^{(i,j)} + \lambda_g^{(i,j)}\right|} \tag{1}$$

where $\lambda_l^{(i,j)}$ and $\lambda_g^{(i,j)}$ are the largest eigenvalues of $\mathbf{C}_l^{(i,j)}$ and $\mathbf{C}_g^{(i,j)}$. They are often referred to as the intensity of the dominant scattering mechanism. This extracted mechanism replaces the original polarization covariance matrix in temporal correlation calculations, streamlining the process by eliminating complex matrix operations [12]. The range of values for $R_{l,g}^{(i,j)}$ is [0, 1]. A value closer to 1 indicates a smaller difference between two polarization covariance matrices. Based on the temporal correlations of any two time points, we can construct the following temporal correlation matrix:

$$\mathbf{Q}^{(i,j)} = \begin{bmatrix} R_{1,1}^{(i,j)} & R_{1,2}^{(i,j)} & \cdots & R_{1,M}^{(i,j)} \\ R_{2,1}^{(i,j)} & R_{2,2}^{(i,j)} & \cdots & R_{2,M}^{(i,j)} \\ \vdots & \vdots & \ddots & \vdots \\ R_{M,1}^{(i,j)} & R_{M,2}^{(i,j)} & \cdots & R_{M,M}^{(i,j)} \end{bmatrix}. \tag{2}$$

By diagonalizing the matrix $\mathbf{Q}(i, j)$, eigenvalues are derived to represent the intensity along each eigenvector direction, specifically denoted as $\tau_1^{(i,j)}, \tau_2^{(i,j)}, \ldots, \tau_k^{(i,j)}, \ldots, \tau_M^{(i,j)}$. The entropy computed from these eigenvalues is then introduced to quantify the degree of disorder of observed land covers over the observation period. Termed as scattering mechanism entropy, it is defined for a pixel (i, j) as follows:

$$H^{(i,j)} = -\sum_{k=1}^{M} S\left(\tau_k^{(i,j)}\right) \cdot \log_M\left[S\left(\tau_k^{(i,j)}\right)\right] \tag{3}$$

where $S(\tau_k^{(i,j)}) = \tau_k^{(i,j)} / \sum_{k=1}^{M} \tau_k^{(i,j)}$, $H^{(i,j)} \in [0, 1]$. A smaller value of $H^{(i,j)}$ signifies reduced correlation differences between PolSAR image pairs at any two time points, reflecting lesser cumulative changes throughout the entire time series. Conversely, as $H^{(i,j)}$ approaches 1, it indicates larger correlation differences between PolSAR image pairs at any two-time points, signifying greater cumulative changes in land cover. In previous work [12], it has been demonstrated that the scattering mechanism entropy for the same land covers follows Gaussian distributions.

2) Optical Feature Extraction: The blue band in the multispectral imagery is highly sensitive to ion concentrations in water [14], which enables effective capture of differences in salt ion concentrations between salt fields and salt lakes. Therefore, the blue band is selected as the optical feature and then fused with the scattering mechanism entropy. Let $B^{(i,j)}$ denote the blue band value of pixel with the coordinate (i, j) in image. Since the scattering mechanism entropy has a range of [0, 1], a normalization operation on $B^{(i,j)}$ is required to adjust the range of $B^{(i,j)}$ to [0, 1]. Besides, Zhang [16] showed that a combination of 2–4 Gaussian distributions effectively models multispectral data distributions. This suggests that blue band data obey Gaussian mixture distributions.

3) Feature Fusion: We fuse scattering mechanism entropy $H^{(i,j)}$ and $B^{(i,j)}$ in a simple and intuitive way as shown in (4) to obtain POSD feature $D^{(i,j)}$

$$D^{(i,j)} = \alpha H^{(i,j)} + \gamma B^{(i,j)} \tag{4}$$

where α and γ are the proportion coefficients of feature fusion, $\alpha,\ \gamma \in [0, 1]$. $D^{(i,j)}$ has information about both the

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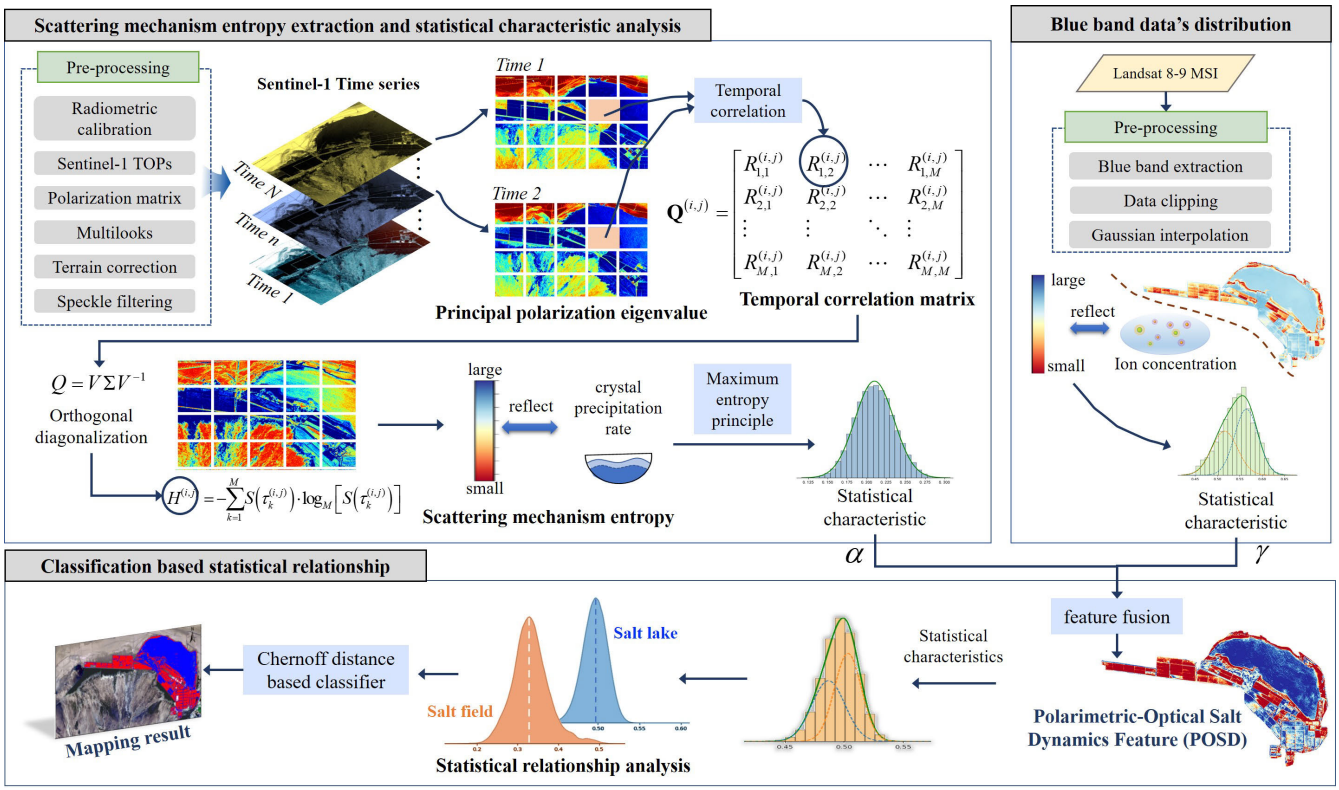


Fig. 2. Flowchart of the hypersaline water body classification by integrating time-series PolSAR and multispectral imagery.

cumulative changes and the ion concentrations in salt lake regions.

The selection of parameters α and γ critically influences the hypersaline water body classification accuracy, as they determine the relative feature contributions in the POSD. We employ a genetic algorithm (GA) [15] to determine the optimal α and γ coefficients, with the following iterative steps.

Step 1: A population of N candidate parameter pairs $\{(\alpha_1, \gamma_1), \ldots, (\alpha_N, \gamma_N)\}$ is randomly generated. The fitness value $\vartheta(\alpha, \gamma)$ for each pair is defined as the overall accuracy (OA) of land-cover classification against ground-truth samples

$$\vartheta(\alpha, \gamma) = \mathrm{OA}\left(D^{(i,j)} = \alpha H^{(i,j)} + \gamma B^{(i,j)}\right) \quad (5)$$

where $\mathrm{OA}(\cdot)$ represents the overall accuracy of the classification results when using $D^{(i,j)}$.

Step 2: Parameter pairs are retained probabilistically through roulette wheel selection, where the selection probability p_k for the kth pair is

$$p_k = \frac{\vartheta(\alpha_k, \gamma_k)}{\sum_{j=1}^{N} \vartheta(\alpha_j, \gamma_j)}. \quad (6)$$

Step 3: Selected parent pairs (α_1, γ_1) and (α_2, γ_2) undergo linear combination to generate offspring $(\alpha_{\mathrm{new1}}, \gamma_{\mathrm{new1}})$ and $(\alpha_{\mathrm{new2}}, \gamma_{\mathrm{new2}})$

$$\begin{cases} \alpha_{\mathrm{new1}} = \upsilon\alpha_1 + (1-\upsilon)\alpha_2, \\ \gamma_{\mathrm{new1}} = \upsilon\gamma_1 + (1-\upsilon)\gamma_2 \\ \alpha_{\mathrm{new2}} = \upsilon\alpha_2 + (1-\upsilon)\alpha_1, \\ \gamma_{\mathrm{new2}} = \upsilon\gamma_2 + (1-\upsilon)\gamma_1 \end{cases} \quad (7)$$

where υ is the crossover point, a random value between 0 and 1.

Step 4: Minor changes are made to the newly generated offspring through mutation operations. Taking $(\alpha_{\mathrm{new1}}, \gamma_{\mathrm{new1}})$ as an example

$$\alpha'_{\mathrm{new1}} = \alpha_{\mathrm{new1}} + \Delta_\alpha, \quad \gamma'_{\mathrm{new1}} = \gamma_{\mathrm{new1}} + \Delta_\gamma \quad (8)$$

where Δ_α and Δ_γ are randomly small amounts of variation, and α'_{new1} and γ'_{new1} are proportion coefficients after the mutation operation.

Steps 2–4 are repeated until the maximum number of iterations is reached or the fitness function reaches the set threshold. The GA-optimized parameters (α^*, γ^*) are utilized to compute the fused feature $D^{(i,j)}$, thereby enhancing the separability between distinct hypersaline water body categories in the classification process.

B. Distribution of POSD

Different land covers typically have POSD probability distributions with different parameters. Therefore, classification of salt lakes and salt fields can be achieved by comparing the similarity of probability distributions. The scattering mechanism entropy has been shown to obey Gaussian distributions [12], while the spectral data has been demonstrated to follow Gaussian mixture distributions [16]. Building on these findings, we can derive the POSD probability distribution.

Let x denote the random variable of the scattering mechanism entropy $H^{(i,j)}$ for the same land cover. Since the scattering mechanism entropy for the same land cover obeys

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Gaussian distributions, we can obtain the probability density function (pdf) $f_{\text{pol}}(x)$ of x

$$f_{\text{pol}}(x) = \frac{1}{\sqrt{2\pi}\sigma_{\text{pol}}} \exp\left[-\frac{(x-\mu_{\text{pol}})^2}{2\sigma_{\text{pol}}^2}\right] \tag{9}$$

where μ_{pol} and σ_{pol} are the mean and standard deviation of the scattering mechanism entropy, respectively. Let y represent the random variable of the blue band $B^{(i,j)}$ in multispectral imagery for the same land cover. To simplify the calculation, we use a combination of two Gaussian distributions to represent the probability distribution of the multispectral data. Thus, the pdf $f_{\text{msi}}(y)$ of y can be expressed as

$$f_{\text{msi}}(y) = \frac{\omega_{\text{msi1}}}{\sqrt{2\pi}\sigma_{\text{msi1}}} \exp\left[-\frac{(y-\mu_{\text{msi1}})^2}{2\sigma_{\text{msi1}}^2}\right] + \frac{\omega_{\text{msi2}}}{\sqrt{2\pi}\sigma_{\text{msi2}}} \exp\left[-\frac{(y-\mu_{\text{msi2}})^2}{2\sigma_{\text{msi2}}^2}\right] \tag{10}$$

where ω_{msi1}, μ_{msi1}, and σ_{msi1} are the weight, mean, and standard deviation of the first Gaussian component, respectively, while ω_{msi2}, μ_{msi2}, and σ_{msi2} are the weight, mean, and standard deviation of the second Gaussian component, respectively. Let z denotes the random variable of the POSD $D^{(i,j)}$, such that $z = \alpha x + \gamma y$. It can be mathematically deduced that the pdf $f(z)$ of z can be expressed as (Appendix A)

$$\begin{aligned} f(z) &= \frac{\omega_{\text{msi1}}}{\sqrt{2\pi}\sqrt{\sigma_{\text{pol}}^2\alpha^2+\gamma^2\sigma_{\text{msi1}}^2}} \exp\left[-\frac{(z-\alpha\mu_{\text{pol}}-\gamma\mu_{\text{msi1}})^2}{2\left(\sigma_{\text{pol}}^2\alpha^2+\gamma^2\sigma_{\text{msi1}}^2\right)}\right] \\ &\quad + \frac{\omega_{\text{msi2}}}{\sqrt{2\pi}\sqrt{\sigma_{\text{pol}}^2\alpha^2+\gamma^2\sigma_{\text{msi2}}^2}} \exp\left[-\frac{(z-\alpha\mu_{\text{pol}}-\gamma\mu_{\text{msi2}})^2}{2\left(\sigma_{\text{pol}}^2\alpha^2+\gamma^2\sigma_{\text{msi2}}^2\right)}\right] \\ &= \frac{\omega_1}{\sqrt{2\pi}\sigma_1} \exp\left[-\frac{(z-\mu_1)^2}{2\sigma_1^2}\right] + \frac{\omega_2}{\sqrt{2\pi}\sigma_2} \exp\left[-\frac{(z-\mu_2)^2}{2\sigma_2^2}\right] \end{aligned} \tag{11}$$

where $\omega_1 = \omega_{\text{msi1}}$, $\omega_2 = \omega_{\text{msi2}}$, $\mu_1 = \alpha\mu_{\text{pol}} + \gamma\mu_{\text{msi1}}$, $\mu_2 = \alpha\mu_{\text{pol}} + \gamma\mu_{\text{msi2}}$, $\sigma_1 = (\sigma_{\text{pol}}^2\alpha^2 + \gamma^2\sigma_{\text{msi1}}^2)^{1/2}$, and $\sigma_2 = (\sigma_{\text{pol}}^2\alpha^2 + \gamma^2\sigma_{\text{msi2}}^2)^{1/2}$. Equation (11) shows that the POSD obeys the Gaussian mixture distribution with two Gaussian components.

C. Chernoff Distance of POSD

Based on the conclusion of the probability distribution above, hypersaline water body classification can be accomplished by quantifying the similarity between Gaussian mixture distributions of different water bodies. Several methods exist for measuring similarity between probability distributions, including the likelihood-ratio test [7], Wasserstein distance [17], and Chernoff distance [18]. PolSAR images typically contain significant noise, which can compromise the accuracy of probability distribution parameter estimation. Methods such as the likelihood-ratio test and Wasserstein distance are particularly sensitive to distribution parameters, potentially leading to substantial calculation errors when parameter estimation is inaccurate. In contrast, the Chernoff distance offers the advantage of parameter independence, meaning it accurately measures the similarity of probability distributions even when the parameter estimation is inaccurate. Given these considerations, we employ the Chernoff distance to quantify the similarity between two Gaussian mixture distributions for land cover classification. We assume that there are two Gaussian mixture distributions with PDFs defined as

$$\begin{cases} f_1(z) = \dfrac{\tilde{\omega}_1}{\sqrt{2\pi}\tilde{\sigma}_1} \exp\left[-\dfrac{(z-\tilde{\mu}_1)^2}{2\tilde{\sigma}_1^2}\right] + \dfrac{\tilde{\omega}_2}{\sqrt{2\pi}\tilde{\sigma}_2} \exp\left[-\dfrac{(z-\tilde{\mu}_2)^2}{2\tilde{\sigma}_2^2}\right] \\ f_2(z) = \dfrac{\bar{\omega}_1}{\sqrt{2\pi}\bar{\sigma}_1} \exp\left[-\dfrac{(z-\bar{\mu}_1)^2}{2\bar{\sigma}_1^2}\right] + \dfrac{\bar{\omega}_2}{\sqrt{2\pi}\bar{\sigma}_2} \exp\left[-\dfrac{(z-\bar{\mu}_2)^2}{2\bar{\sigma}_2^2}\right] \end{cases} \tag{12}$$

where $\tilde{\omega}_1$, $\tilde{\mu}_1$, $\tilde{\sigma}_1$, $\bar{\omega}_1$, $\bar{\mu}_1$, and $\bar{\sigma}_1$ are the weights, means, and standard deviations of the first Gaussian component. $\tilde{\omega}_2$, $\tilde{\mu}_2$, $\tilde{\sigma}_2$, $\bar{\omega}_2$, $\bar{\mu}_2$, and $\bar{\sigma}_2$ are the weights, means, and standard deviations of the second Gaussian component. For two PDFs, Duda et al. [19] gave the definition of Chernoff distance $D(f_1, f_2)$

$$D(f_1, f_2) = -\ln\left[\int f_1^{\beta}(z) f_2^{1-\beta}(z) dz\right] \tag{13}$$

where β is a parameter. Substituting (12) into (13), we can derive the Chernoff distance between two Gaussian mixture distributions (Appendix B). The Chernoff distance for two Gaussian mixture distributions is

$$\begin{aligned} D(f_1, f_2) &= \ln\left(\tilde{\omega}_1+\tilde{\omega}_2\right)\left(\bar{\omega}_1+\bar{\omega}_2\right) \\ &\quad - \ln\Bigg\{ \frac{\tilde{\omega}_1^{\beta}\bar{\omega}_1^{(1-\beta)}}{\sqrt{2\pi\left(\tilde{\sigma}_1^2/\beta+\bar{\sigma}_1^2/1-\beta\right)}} \times \exp\left[-\frac{\left(\tilde{\mu}_1-\bar{\mu}_1\right)^2}{2\left(\tilde{\sigma}_1^2/\beta+\bar{\sigma}_1^2/1-\beta\right)}\right] \\ &\quad + \frac{\tilde{\omega}_1^{\beta}\bar{\omega}_2^{(1-\beta)}}{\sqrt{2\pi\left(\tilde{\sigma}_1^2/\beta+\bar{\sigma}_2^2/1-\beta\right)}} \times \exp\left[-\frac{\left(\tilde{\mu}_1-\bar{\mu}_2\right)^2}{2\left(\tilde{\sigma}_1^2/\beta+\bar{\sigma}_2^2/1-\beta\right)}\right] \\ &\quad + \frac{\tilde{\omega}_2^{\beta}\bar{\omega}_1^{(1-\beta)}}{\sqrt{2\pi\left(\tilde{\sigma}_2^2/\beta+\bar{\sigma}_1^2/1-\beta\right)}} \times \exp\left[-\frac{\left(\tilde{\mu}_2-\bar{\mu}_1\right)^2}{2\left(\tilde{\sigma}_2^2/\beta+\bar{\sigma}_1^2/1-\beta\right)}\right] \\ &\quad + \frac{\tilde{\omega}_2^{\beta}\bar{\omega}_2^{(1-\beta)}}{\sqrt{2\pi\left(\tilde{\sigma}_2^2/\beta+\bar{\sigma}_2^2/1-\beta\right)}} \end{aligned}$$

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$$\times \exp\left[-\frac{\left(\tilde{\mu}_2-\bar{\mu}_2\right)^2}{2\left(\tilde{\sigma}_2^2/\beta+\bar{\sigma}_2^2/1-\beta\right)}\right]\Bigg\}. \tag{14}$$

Notably, Duda et al. [19] proved that for any probability distribution, the Chernoff distance best describes the similarity between two probability distributions when $\beta = 0.66$. Therefore, in subsequent experiments, $\beta = 0.66$ is set to classify the salt lake region. Equation (14) can be used to calculate the similarity of two Gaussian mixture distributions. The more similar the two Gaussian mixture distributions are, the smaller $D(f_1, f_2)$ will be.

D. Hypersaline Water Body Classification via Chernoff Distance

Equation (11) can be utilized to compute the Chernoff distance of POSD between unknown and known hypersaline water body types. Unknown hypersaline water body types are then classified according to the minimum Chernoff distance. This calculation requires random samples from the unknown hypersaline water body to estimate its probability distribution parameters. However, a typical PolSAR scene encompasses multiple land cover types with distinct statistical properties [20]. Arbitrary pixel selection may result in samples from different land covers, leading to inaccurate parameter estimation. To mitigate this issue, we employ a superpixel-based sample selection approach. A superpixel is a set of neighboring pixels with similar characteristics, and all pixels within a superpixel can be considered to belong to the same land cover. Thus, we segment a PolSAR scene into multiple superpixels, and all POSD within a superpixel are treated as a set of samples, ensuring more accurate parameter estimation. In our method, the Watershed algorithm [21] is employed to generate superpixels. This algorithm effectively handles irregular-shaped objects by identifying regions based on gradients or watershed lines, enabling accurate segmentation of objects with complex shapes or fragmented boundaries.

The PolSAR scene is segmented into multiple superpixels using the Watershed algorithm, with $\hat{Z} = \{z_i\}_{1\le i\le \kappa}$ denoting an individual superpixels, where z_i is the POSD of the $(1 \le i \le \kappa)$ pixel. To calculate the Chernoff distance, the expectation maximization (EM) algorithm [22] is used to estimate the parameters of the Gaussian mixture distribution within each superpixel. We model the Gaussian mixture distribution with two Gaussian components, characterized by parameters $\hat{\mu}_1$, $\hat{\mu}_2$, $\hat{\omega}_1$, $\hat{\omega}_2$, $\hat{\sigma}_1$, and $\hat{\sigma}_2$, where $\hat{\omega}_1$, $\hat{\mu}_1$, and $\hat{\sigma}_1$ are the weight, mean, and standard deviation of the first Gaussian component, respectively. $\hat{\omega}_2$, $\hat{\mu}_2$, and $\hat{\sigma}_2$ denote these parameters for the second Gaussian component. The EM algorithm performs parameter estimation as follows.

Step 1: Initialize the parameters of the Gaussian mixture distribution, that is,

$$\begin{cases} \hat{\mu}_1 = z_1, \quad \hat{\mu}_2 = z_2, \quad \hat{\omega}_1 = 0.5, \quad \hat{\omega}_2 = 0.5 \\ \hat{\sigma}_1, \hat{\sigma}_2 = \dfrac{1}{\kappa}\displaystyle\sum_{i=1}^{\kappa}\left(z_i - \frac{1}{\kappa}\sum_{j=1}^{\kappa} z_i\right)^2. \end{cases} \tag{15}$$

Step 2: Execute the expectation step to compute the parameter $\hat{\gamma}_i$, that is,

$$\hat{\gamma}_i = \frac{\hat{\omega}_1\phi_1(z_i)}{\hat{\omega}_2\phi_1(z_i)+\hat{\omega}_1\phi_2(z_i)} \tag{16}$$

where $\phi_1(z_i) = 1/(2\pi)^{1/2}\hat{\sigma}_1 \cdot \exp[-(z_i-\hat{\mu}_1)^2/2\hat{\sigma}_1^2]$, $\phi_2(z_i) = 1/(2\pi)^{1/2}\hat{\sigma}_2 \cdot \exp[-(z_i-\hat{\mu}_2)^2/2\hat{\sigma}_2^2]$ and $i = 1, 2, \ldots, \kappa$.

Step 3: Perform the maximization step to update the mean, variance, and weights

$$\begin{cases} \hat{\mu}_1 = \dfrac{\sum_{i=1}^{\kappa}\left(1-\hat{\gamma}_i\right)z_i}{\sum_{i=1}^{\kappa}\left(1-\hat{\gamma}_i\right)}, \quad \hat{\mu}_2 = \dfrac{\sum_{i=1}^{\kappa}\hat{\gamma}_i z_i}{\sum_{i=1}^{\kappa}\hat{\gamma}_i} \\ \hat{\sigma}_1^2 = \dfrac{\sum_{i=1}^{\kappa}\left(1-\hat{\gamma}_i\right)\left(z_i-\hat{\mu}_1\right)^2}{\sum_{i=1}^{\kappa}\left(1-\hat{\gamma}_i\right)} \\ \hat{\sigma}_2^2 = \dfrac{\sum_{i=1}^{\kappa}\hat{\gamma}_i\left(z_i-\hat{\mu}_2\right)^2}{\sum_{i=1}^{\kappa}\hat{\gamma}_i} \\ \hat{\omega}_1 = \displaystyle\sum_{i=1}^{\kappa}\hat{\gamma}_i/N, \quad \hat{\omega}_2 = 1-\sum_{i=1}^{\kappa}\hat{\gamma}_i/N. \end{cases} \tag{17}$$

The expectation step and maximization step are repeated until convergence. The resulting $\hat{\mu}_1$, $\hat{\mu}_2$, $\hat{\omega}_1$, $\hat{\omega}_2$, $\hat{\sigma}_1$, and $\hat{\sigma}_2$ are the final parameters of the Gaussian mixture distribution.

The aforementioned procedure enables estimation of Gaussian mixture distribution parameters for all POSD within a superpixel. Analogously, the EM algorithm is applied to compute parameters for known hypersaline water body types. By substituting these parameter estimates—both for superpixels and known hypersaline water body types—into (11), we can quantify the similarity between each superpixel and known hypersaline water body categories. Classification is then achieved by assigning each superpixel to the hypersaline water body type, yielding the minimum Chernoff distance. This approach leverages superpixels for parameter estimation, effectively utilizing spatial neighborhood information and mitigating the impact of noise on classification outcomes.

III. Study Area and Data

A. Study Area

This study focuses on the four largest salt lake sites in China, as shown in Fig. 3. These lakes play a vital role in maintaining the balance between water and salt in the ecosystem and are key salt production bases [23]. Thus, classifying these areas is both ecologically and industrially significant. Detailed descriptions of the four salt lakes are provided below.

1) *Qarhan Salt Lake:* Qarhan salt lake is located in Qinghai Province, China, with geographic coordinates of 94°14′–94°15′ East longitude and 36°37′–37°12′ North latitude. As the largest salt lake in China, Qarhan salt lake is renowned for its rich mineral resources, including potassium, magnesium, sodium, lithium, boron, and other salt mine resources [24].
2) *Yiliping Salt Lake:* Yiliping salt lake is located in the central part of the Qaidam Basin in China, with longitude

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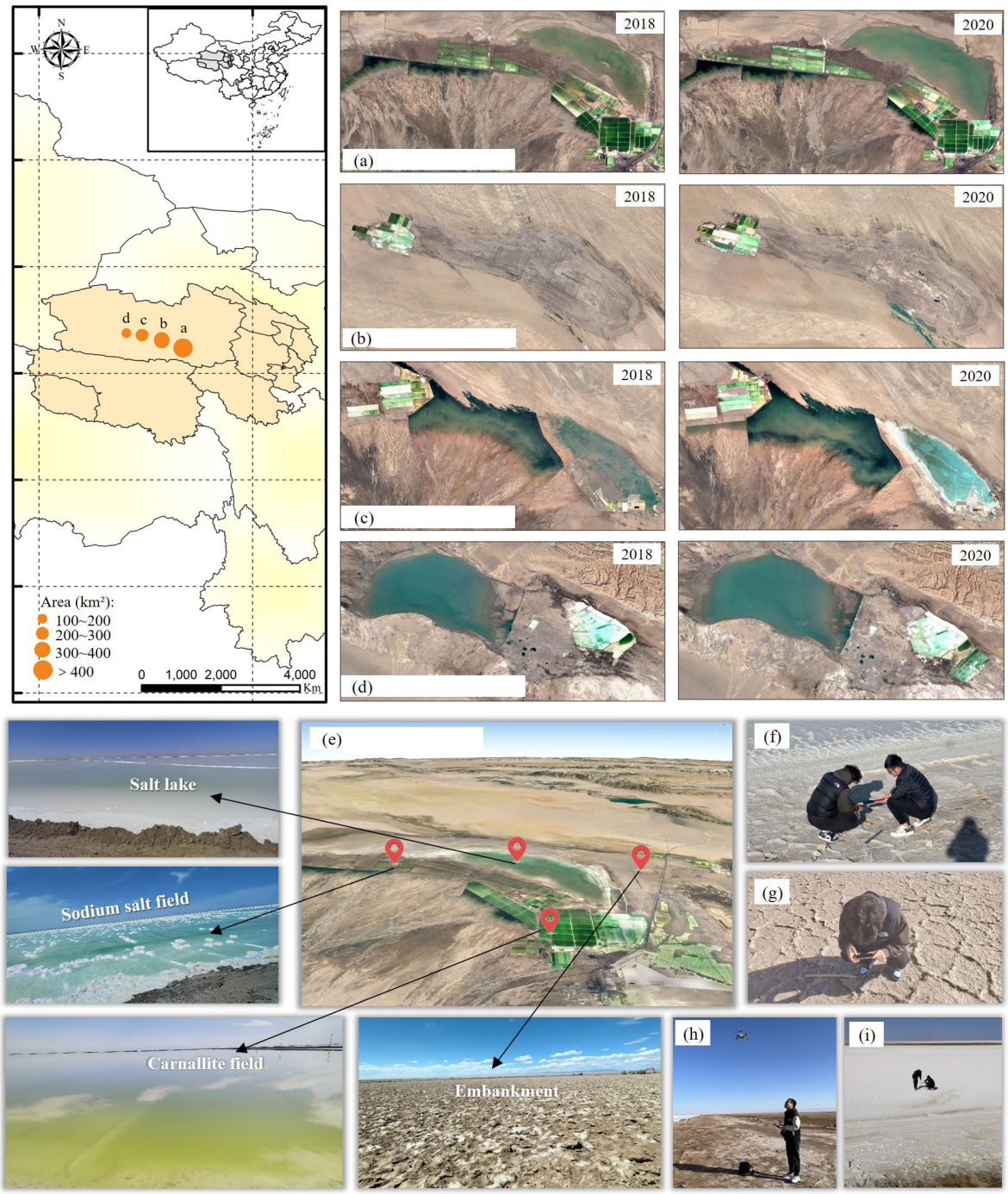


Fig. 3. Overview of the study area. (a)–(d) Location and optical image of the four study areas. (e) Various land cover types in the Qarhan salt lake. (f)–(i) Photographs of the field survey.

92°58′–93°20′ East and latitude 37°51′–38°04′ North. It has a brine reserve of 985 million cubic meters, with lithium chloride reserves of 1.897 million tons and potassium chloride reserves of 18.659 million tons [25].

3) *Taijnar Salt Lake:* The Taijnar salt lake consists of two parts: East Taijnar salt lake and West Taijnar salt lake. Taijnar salt lake is a comprehensive super-large brine deposit in China, containing abundant mineral resources such as lithium, boron, potassium, and others [26].
4) *Gasikule Salt Lake:* This closed endorheic basin, located between 90°40′–91°10′ East longitude and 37°57′–38°11′ North latitude, receives perennial recharge from the Aral and Timurik Rivers. It is characterized as an extensive rock salt brine deposit with coexisting solid-liquid phases [27].

B. Data and Preprocessing

In this study, two types of data sources—PolSAR data and multispectral images—are utilized to construct the dataset. The details of the dataset for each study area are shown in Table I. For PolSAR data, we employ Sentinel-1A data acquired in

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TABLE I
DETAILS OF THE DATASET FOR EACH STUDY AREA

Study area	PolSAR data		Multispectral data	
	Satellite	Acquisition dates (YY/MM/DD)	Satellite	Acquisition dates (YY/MM/DD)
Qarhan	Sentinel-1A	2022/01/20; 2022/02/13; 2022/03/21; 2022/04/14; 2022/05/08; 2022/06/13.	Landsat-8	2022/07/19
Yiliping	Sentinel-1A	2023/01/25; 2023/02/18; 2023/03/26; 2023/04/19; 2023/05/25; 2023/06/18.	Landsat-9	2023/01/01
Taijnar	Sentinel-1A	2023/01/25; 2023/02/18; 2023/03/26; 2023/04/19; 2023/05/25; 2023/06/18.	Landsat-9	2023/08/06
Gasikule	Sentinel-1A	2023/03/20; 2023/04/25; 2023/05/19; 2023/06/24.	Landsat-8	2023/07/26

early 2022 and 2023. Each Sentinel-1 satellite operates on a 12-day repeat cycle, which meets the need for continuous observation of land cover [28].

We employ the Sentinel Application Platform (SNAP, version 9.0) and the Polarimetric SAR Processing and Educational Tool (PolSARpro, version 6.0) for data preprocessing tasks. Initially, we calibrate the acquired SLC Sentinel-1 data to ensure that the radar signal accurately represents the actual area visible to the radar. Subsequently, the Sentinel-1 TOPS operators are used to implement the swath, polarization selection, and the deburst of Sentinel-1 data. The Sentinel-1 images are then converted into polarization covariance matrices and multilooked to reduce speckle noise. Finally, all polarization covariance matrices undergo range–Doppler terrain correction to align all pixels with their corresponding ground positions. And the filtering is applied to these data to further mitigate the effects of speckle noise [29].

The multispectral data utilized in this study are acquired by Landsat 8 and Landsat 9 (Table I), providing data across 11 spectral bands. The Landsat 8 and 9 data are Level-2 products that have undergone radiometric calibration, geometric correction, and atmospheric correction [30]. Therefore, no additional preprocessing is required except for resolution adjustment, which aligns the pixels of Landsat 8 and 9 data with PolSAR data. To minimize the impact on data distribution during this adjustment, we employed Gaussian process interpolation [31]. This method models neighboring pixels as Gaussian distributions, adding new pixels in a way that preserves the original data distribution as much as possible.

IV. EXPERIMENTAL RESULTS

A. Feasibility Analysis of Classification Methods

To validate the proposed method and corroborate our conclusions, we first calculate the scattering mechanism entropy for each study area using Sentinel-1 data. The results are illustrated in Fig. 4. Taking Qarhan salt lake as an example, regions A, B, and C correspond to salt lakes, sodium salt fields, and carnallite fields, respectively. A notable disparity in scattering mechanism entropy is evident between salt lakes and carnallite fields. Salt lakes exhibit larger cumulative changes, resulting in higher scattering mechanism entropy, predominantly ranging from 0.2 to 0.25. Conversely, carnallite fields display smaller cumulative changes, with scattering mechanism entropy primarily concentrated between 0.05 and 0.1. While salt lakes and sodium salt fields show nearly identical scattering mechanism entropy due to similar cumulative changes, rendering this feature alone insufficient for their differentiation. Next, we extract the blue band from the multispectral data, as depicted in Fig. 4. There is a distinct difference in the blue band values between the sodium salt fields and the salt lakes, with a difference of about 0.1. This indicates the blue band's efficacy in distinguishing sodium salt fields from salt lakes. Note that the blue band alone is inadequate for differentiating between salt lakes and carnallite fields due to their similar salt ion concentrations. Finally, we calculate the POSD by fusing the scattering mechanism entropy and the blue band. Fig. 5 depicts the pdf curves of POSD features for each study area. It demonstrates that POSD effectively captures the cumulative changes and ion concentrations in salt lake regions, resulting in discernible differences between salt lakes and salt fields. Although POSD intervals for salt lakes and salt fields in some study areas overlap, the extent of concentration of POSDs differs between these hypersaline water body types. The POSDs of salt lakes and salt fields are concentrated around 0.45 and 0.35, respectively, providing crucial information for salt lake region classification.

In each study area, we select a number of salt lakes' POSD for data fitting to verify the distribution that each feature follows. A normal distribution is used to fit the scattering mechanism entropy, while Gaussian mixture distributions with two Gaussian components are employed to fit the blue band and the POSD. Fig. 6 illustrates the fitting results, showing excellent agreement between the probability distribution curves and the features' histograms. This confirms the accuracy of the conclusions presented in Section II. In addition, we also select a portion of the salt fields' POSD to be fit with Gaussian mixture distributions. The comparative results of the salt lake and salt field POSD fitting are shown in Fig. 7. The fitting results illustrate that the POSD of both salt lakes and salt fields obey Gaussian mixture distributions, and the two distributions are significantly different in terms of parameters and distribution intervals. Therefore, we can utilize the differences in probability distributions to classify salt lakes and salt fields.

B. Classification Evaluations

To evaluate the validity of the proposed method, seven land-cover classification methods are selected for performance comparison: time-series change analysis classification (TSCA) [32], spatiotemporal multilevel attention classification (STMA) [33], CNN classification based on multispectral data (CNNM) [34], integration of optical and PolSAR features classification (IOPF) [35], classification of integration multitemporal PolSAR and multispectral data based on extreme gradient boosting (MMXG) [36], classification based on Bhattacharyya distance (Bhat) [37], and classification based on minimum distance (MinD). TSCA first obtains a new time-series change feature through generalized eigen decomposition, which quantifies the magnitude of variation between PolSAR images acquired at two different times. Afterward, the new time-series change feature is input to the Random Forest classifier for

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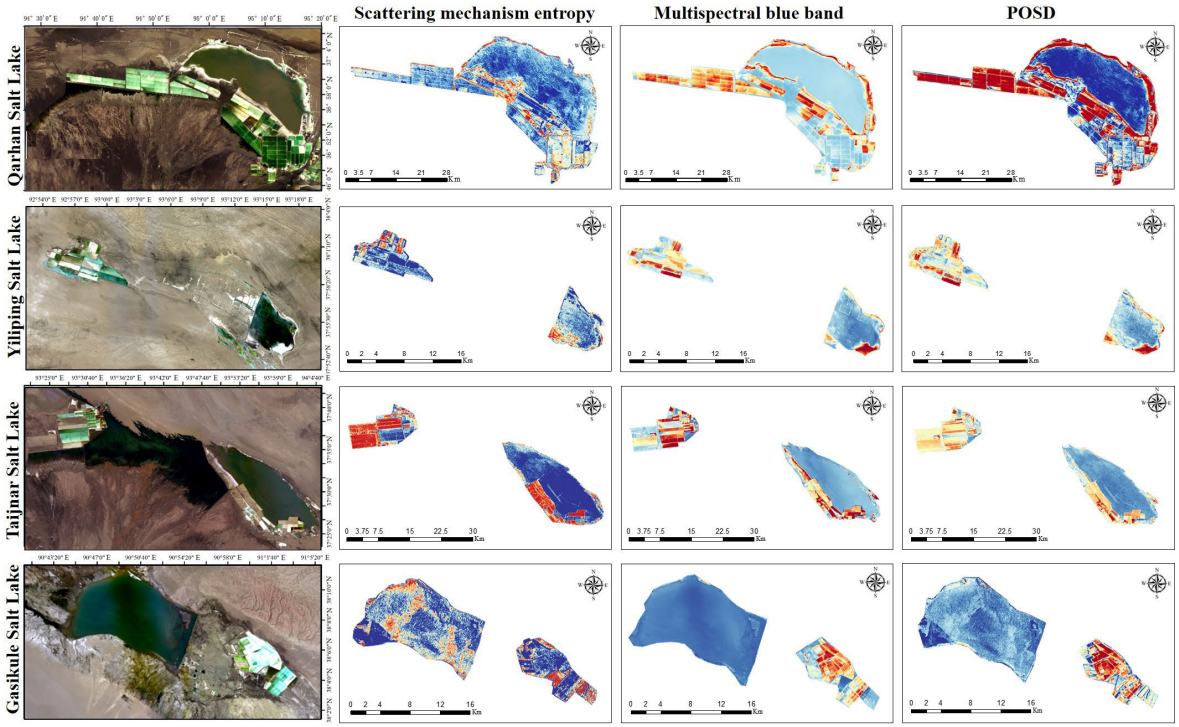


Fig. 4. Scattering mechanism entropy, multispectral blue band, and POSD for Qarhan salt lake, Yiliping salt lake, Taijnar salt lake, and Gasikule salt lake.

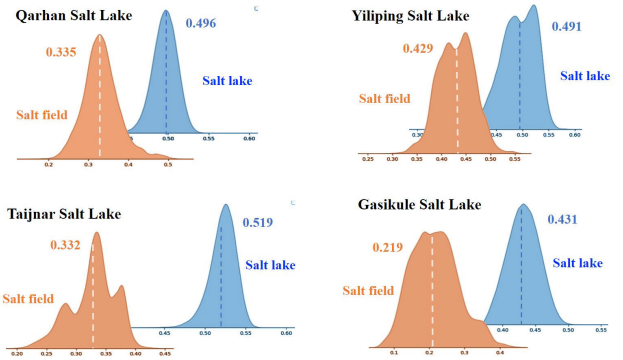


Fig. 5. PDF curves of POSD features for Qarhan salt lake, Yiliping salt lake, Taijnar salt lake, and Gasikule salt lake.

land-cover classification. STMA can analyze the multiscale spatiotemporal correlation between any two SAR images in a time series by a multilevel attention mechanism. Besides, to ensure a fine extraction of multigranularity features, a learnable spatial attention position encoding is used to adaptively generate the position priors. In CNNM, the convolutional neural network is improved for better capturing the spectral information of different land covers to obtain highly accurate classification results. IOPF first integrates optical and PolSAR features into a high-dimensional feature matrix. Afterward, this feature matrix is input into a support vector machine classifier for land-cover classification. MMXG is a land-cover classification method based on multitemporal PolSAR and multispectral data. The method first extracts polarization features from each PolSAR image in the time series and then inputs them together with spectral features into an improved extreme gradient boosting classifier for classification. Bhat and MinD are comparison experiments set up to validate the performance of the proposed Chernoff distance-based classifier. These two methods input POSD to the Bhattacharyya distance classifier and the minimum distance classifier for salt lake classification, respectively. These comparison methods involve multitemporal PolSAR, multispectral, and integration of PolSAR and optical features land-cover classification methods, enabling a comprehensive evaluation of the effectiveness of the proposed methods.

All methods are applied to map the classification results for the four study areas. The input to TSCA is the polarization covariance matrix of the Sentinel-1 data acquired at two different time points, and the number of decision trees for the random forest classifier is 100. In STMA, a time series of Sentinel-1 data acquired at different times is input to the network, which is trained for 100 epochs with a learning rate of 0.001. For CNNM, the blue band from the multispectral data served as input for land-cover classification. The network is also trained for 100 epochs with a learning rate of 0.001. In IOPF, the high-dimensional feature matrix is constructed using polarization scattering entropy, scattering angle, and blue, red, and green bands from multispectral data. The support vector machine classifier then employs a hinge loss function with a maximum of 100 iterations. In MMXG, the polarization scattering entropy, scattering angle, Span, intensity of each polarization channel for each PolSAR image, as well as the red band, blue band, green band, and Modified Normalized Difference Water Index from the multispectral data are input to the classifier. The number of decision trees for the extreme gradient boosting classifier is

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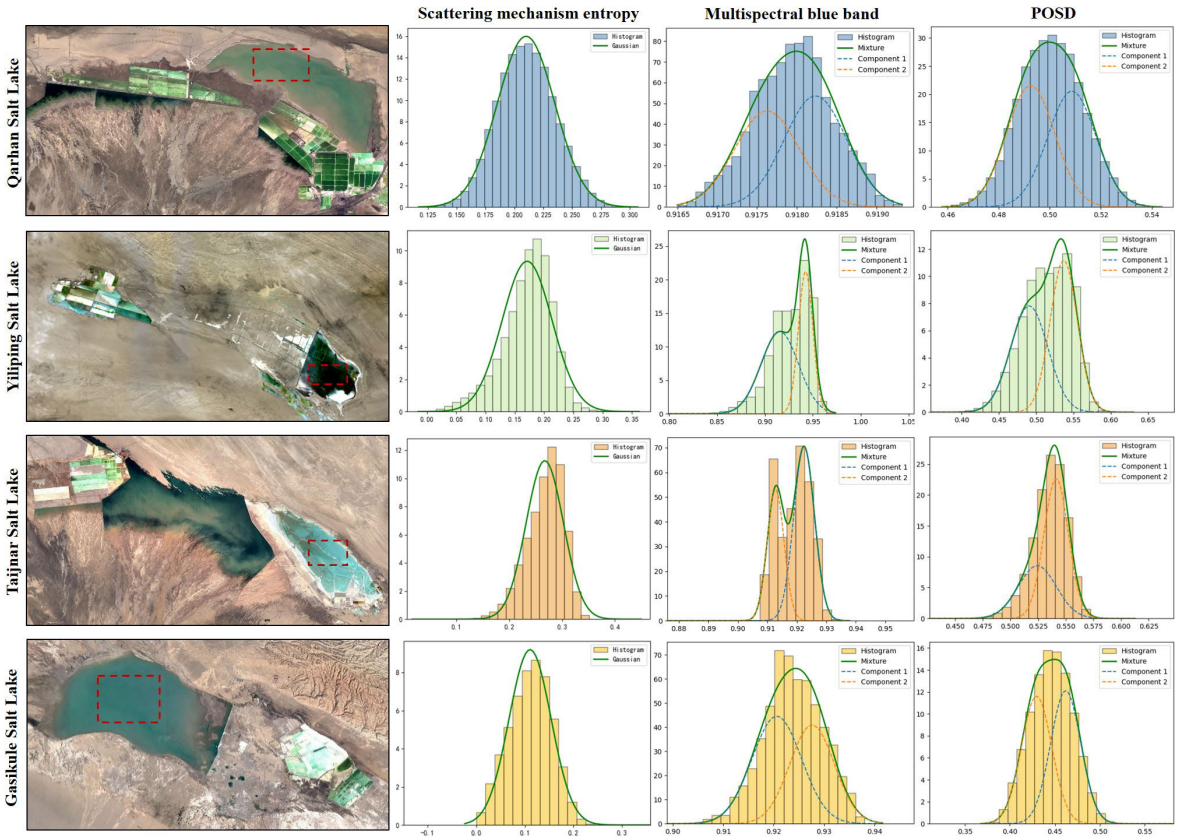


Fig. 6. Fitting results of scattering mechanism entropy, multispectral blue band, and POSD for four salt lake regions.

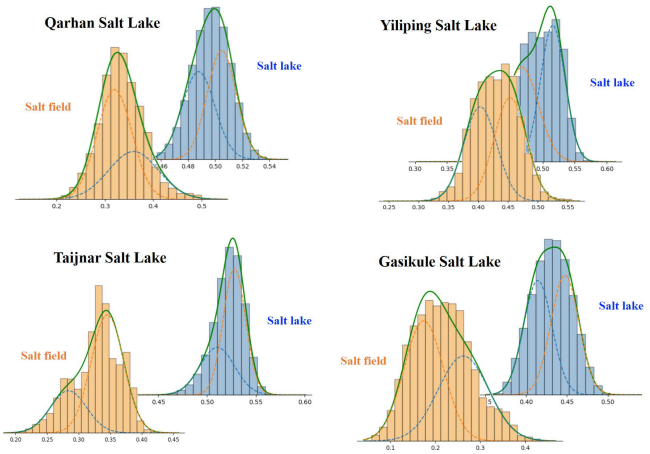


Fig. 7. Fitting results of Gaussian mixture distributions for salt lakes and salt fields' POSD in the four study areas.

set to 100. Fig. 8 shows a visual comparison of the land-cover classification results generated by all methods on the four datasets.

Fig. 8 demonstrates that the classification results of the proposed method align closely with the ground-truth image, showcasing superior noise resistance and spatial consistency. This can be attributed to the effectiveness of POSD in capturing cumulative changes and ion concentrations in salt lake regions, thereby amplifying the distinctions between salt lakes and salt fields. Moreover, the implementation of a Chernoff distance-based classification method using superpixels as processing units maximizes the utilization of neighborhood information. In contrast, the comparative methods show various limitations. The time-series change feature in TSCA can only reflect land-cover changes between two specific time points, potentially failing to accurately represent changes throughout the salt production cycle in the salt lake region. STMA obtains the change of the whole time series by analyzing the spatio-temporal correlation of any two PolSAR images across the entire time series. Compared with TSCA, STMA can better represent the changes of salt lakes and salt fields, so the visual effect of classification is better. However, STMA still exhibits a substantial number of misclassified pixels. CNNM relies solely on optical features for classification, but struggles with distinguishing between salt lakes and fields due to similar salt ion concentrations on the carnallite fields and salt lakes. Although IOPF integrates optical and polarization features, its inability to capture information about changes in land cover results in classification results inferior to the proposed method. Similarly, MMXG simply stacks all polarization and spectral features and does not reveal the relationships between the features. As for Bhat and MinD, although their inputs are both POSD, their classification is performed by calculating the interclass distance between pixels and samples. The process of calculating the distance is very sensitive to noise. In addition, if the number of samples is not balanced, these classifiers will make a judgment more in favor of the category with the larger number of samples. Notably, all comparative methods, except our proposed approach, use individual pixels as the processing unit. This

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Fig. 8. Classification results of multiple methods for the four salt lake regions in China.

leads to the occurrence of numerous isolated false pixels in the classification results, which are challenging to eliminate through postprocessing steps (e.g., morphological processing). In summary, the results of the proposed method yield a smoother visual effect and fewer misdetections than the compared methods.

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TABLE II
QUANTITATIVE EVALUATION RESULTS OF ALL METHODS ON THE FOUR STUDY AREAS

Study area	method	*Pc*(%)		*R*(%)		*F*1(%)		*OA*(%)	*Kappa*(%)
		Lake	Field	Lake	Field	Lake	Field		
Qarhan	Proposed	**88.76**	94.00	95.48	**85.39**	**92.00**	**89.49**	**90.91**	**81.52**
	TSCA	63.70	96.60	98.68	39.95	77.42	56.53	70.28	39.37
	STMA	84.19	**99.96**	**99.98**	77.32	91.41	87.20	89.72	78.85
	CNNM	72.80	99.94	99.97	54.88	84.25	70.85	79.55	57.07
	IOPF	73.25	**99.96**	99.96	55.89	84.56	71.70	80.02	58.09
	MMXG	79.74	99.58	99.76	69.39	88.63	81.79	86.01	71.00
	Bhat	88.68	85.97	88.32	86.39	88.50	86.18	87.45	74.68
	MinD	76.48	99.89	99.94	62.89	86.65	77.18	83.16	64.89
Yiliping	Proposed	**98.90**	86.35	86.54	**99.25**	**92.78**	**92.68**	**92.73**	**85.53**
	TSCA	80.18	75.39	78.53	77.22	79.34	76.29	77.92	55.64
	STMA	79.68	**99.96**	**99.96**	70.04	88.67	82.37	86.22	71.62
	CNNM	79.87	80.75	84.79	74.92	82.26	77.73	80.25	60.03
	IOPF	81.16	81.73	85.39	76.73	83.22	79.15	81.41	62.40
	MMXG	93.03	87.33	88.60	92.20	90.76	89.70	90.26	80.47
	Bhat	80.67	67.28	66.27	81.37	72.78	73.66	73.23	46.93
	MinD	71.99	74.72	81.97	62.56	76.66	68.10	73.04	45.07
Taijnar	Proposed	**97.41**	97.35	99.36	**89.95**	**98.37**	**93.51**	**97.40**	**91.88**
	TSCA	90.80	82.52	96.50	62.80	93.57	71.32	89.49	65.03
	STMA	91.67	**99.80**	**99.96**	65.33	95.64	78.97	92.77	74.84
	CNNM	90.67	78.39	95.46	62.64	93.01	69.63	88.63	62.75
	IOPF	93.56	71.14	91.90	75.93	92.72	73.45	88.58	66.19
	MMXG	92.46	86.73	97.19	69.84	94.77	77.37	91.50	72.21
	Bhat	96.83	67.88	88.94	88.94	92.72	76.99	88.94	69.88
	MinD	93.52	77.68	94.33	75.12	93.92	76.38	90.33	70.30
Gasikule	Proposed	**97.66**	**99.98**	99.52	**88.51**	**98.60**	**93.89**	**97.65**	**91.36**
	TSCA	87.23	79.16	98.40	29.61	92.48	43.10	86.71	37.30
	STMA	88.68	99.80	**99.98**	37.67	94.01	54.72	89.41	50.09
	CNNM	95.80	97.72	99.62	78.68	97.68	87.17	96.07	84.88
	IOPF	94.87	96.79	99.49	73.72	97.13	83.69	95.12	80.88
	MMXG	95.62	85.62	97.31	78.25	96.46	81.77	94.07	78.24
	Bhat	96.18	71.62	93.35	81.91	94.75	76.42	91.41	71.19
	MinD	93.09	96.49	99.52	63.94	96.20	76.91	93.48	73.29

We calculate precision (Pc), recall (R), $F1$-score ($F1$), OA, and kappa coefficient (Kappa) of the classification results for the four study areas to quantitatively evaluate the validity of the mapping results for the salt lake region. The quantitative analysis presented in Table II demonstrates the superior performance of our proposed method across most evaluation metrics except for R for salt lakes and Pc for salt fields. This pattern can be attributed to the tendency of alternative methods to misclassify salt fields as salt lakes, resulting in marginally higher Pc for salt lakes and R for salt fields in their results. However, our method exhibits significant superiority across other crucial evaluation metrics, particularly in Kappa and OA. These metrics provide a balanced measure of classification performance. Besides, the markedly higher $F1$ and OA values indicate our method's results are closest to real land cover, corroborating the visual observations discussed earlier. This consistent outperformance across multiple metrics suggests enhanced stability and efficacy of our approach compared to the alternatives evaluated. The comprehensive superiority demonstrated by our method, despite slight underperformance in two specific metrics, provides strong evidence for its effectiveness in high-quality salt lake region mapping.

C. Results of the Salt Lake Region Classification Mapping

The proposed method is applied to classify salt lakes and salt fields across four study areas, as depicted in Fig. 9, covering the past three years. These results form a reliable foundation for analyzing the spatial distribution and temporal changes of land covers in these regions. Regarding spatial distribution, the Qarhan salt lake region is abundant in salt fields, densely concentrated in its southern part. These fields are evenly distributed, positioned near salt lakes to facilitate efficient brine extraction. In the Yiliping salt lake region, salt fields are primarily located in the western area, whereas salt lakes are concentrated in the eastern area, with a considerable distance between them. In addition, there are some abandoned or dormant salt fields in the central part of the region, and the salt lakes were also dry in 2022 and 2021. The Taijnar salt lake, comprising West Taijnar and East Taijnar, is located in the western and eastern parts of the region, respectively. The West Taijnar region includes numerous salt fields, some of which have dried up and transformed into land formations. Similarly, a significant portion of the salt fields in the East Taijnar region has also dried up, leaving only a small fraction intact. The intermediate region in the Taijnar salt lake lies in a

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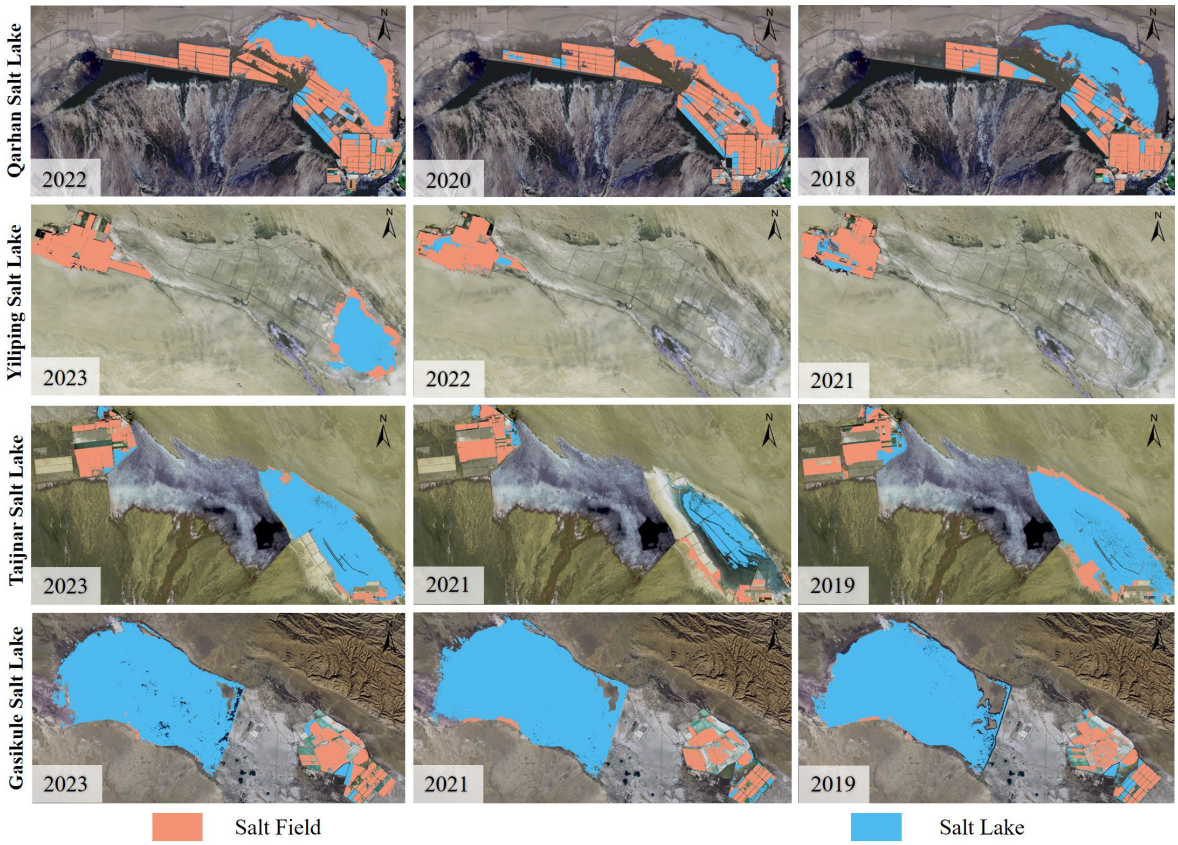


Fig. 9. Three years of mapping results for the four largest salt lake regions in China.

region of salt lakes, with the section near West Taijnar featuring the Water Yadan, a renowned scenic attraction in China. The distribution of salt fields and salt lakes in the Gasikule salt lake region is more balanced. Salt lakes predominantly occupy the western region, while salt fields are concentrated in the eastern region. Furthermore, only a small portion of the salt fields have dried up, with most remaining actively used.

To better observe the temporal changes in salt lakes and salt fields within each study area, we select two subregions from each area and present the classification results in Fig. 10. A comparative analysis of Fig. 10(a) and (b) reveals an increase in salt fields of the Qarhan salt lake region in 2022. Furthermore, Fig. 10(c) compared with 10(b) shows a noticeable expansion in the salt lakes area in 2020. Fig. 10(d)–(f) shows the variations in carnallite fields within the Qarhan salt lake region, illustrating that the extent of carnallite fields has remained largely constant over the years. In the Yiliping salt lake, Fig. 10(g)–(i) indicates that the area of salt fields in the northern part increased in 2022 and then decreased again by 2023. Meanwhile, Fig. 10(j)–(l) shows changes in the southeastern part of the Yiliping salt lake region. New salt fields emerged in the southeast in 2022, disappeared by 2023, and then reappeared with another new salt field. For the western part of the Taijnar salt lake region, a portion of the salt fields disappeared by 2021, with new salt fields emerging by 2023, as shown in Fig. 10(m)–(o). Fig. 10(p)–(r) highlights the changes in the eastern part of the Taijnar salt lake region, where the distribution of salt fields varied each year, and 2021 saw widespread drying up of the salt lakes. In the Gasikule salt lake, the area of the salt lakes is increasing every year, and the changes in the distribution of salt fields can be seen in Fig. 10(v)–(x). These findings illuminate the spatial patterns and temporal dynamics of salt lakes and salt fields across the study areas, providing valuable insights for land management and environmental monitoring in these ecologically sensitive regions.

D. Classification of Sodium Salt and Carnallite Fields

Theoretically, the salt field regions can be finely divided into two categories, with samples of sodium salt and carnallite fields. In the case of Qarhan salt lake, we subdivided the salt field region into two categories, thereby classifying Qarhan salt lake into three distinct categories: salt lakes, sodium salt fields, and carnallite fields. The classification results, along with radar charts displaying evaluation indicators, are presented in Fig. 11, where R_1, R_2, and R_3 are recall for salt lakes, sodium salt fields, and carnallite fields, respectively. MR is the mean recall, that is, the average of R_1, R_2, and R_3.

It can be seen that from Fig. 11(b) the sodium salt fields are mainly concentrated in the northwestern region, while the carnallite fields are mainly located in the southeastern region. Each category of salt fields is concentrated for easier management. Regarding evaluation indicators, both salt lakes and sodium salt fields exhibit recall exceeding 80%. However, the recall for carnallite fields is only 50%. Upon comparing

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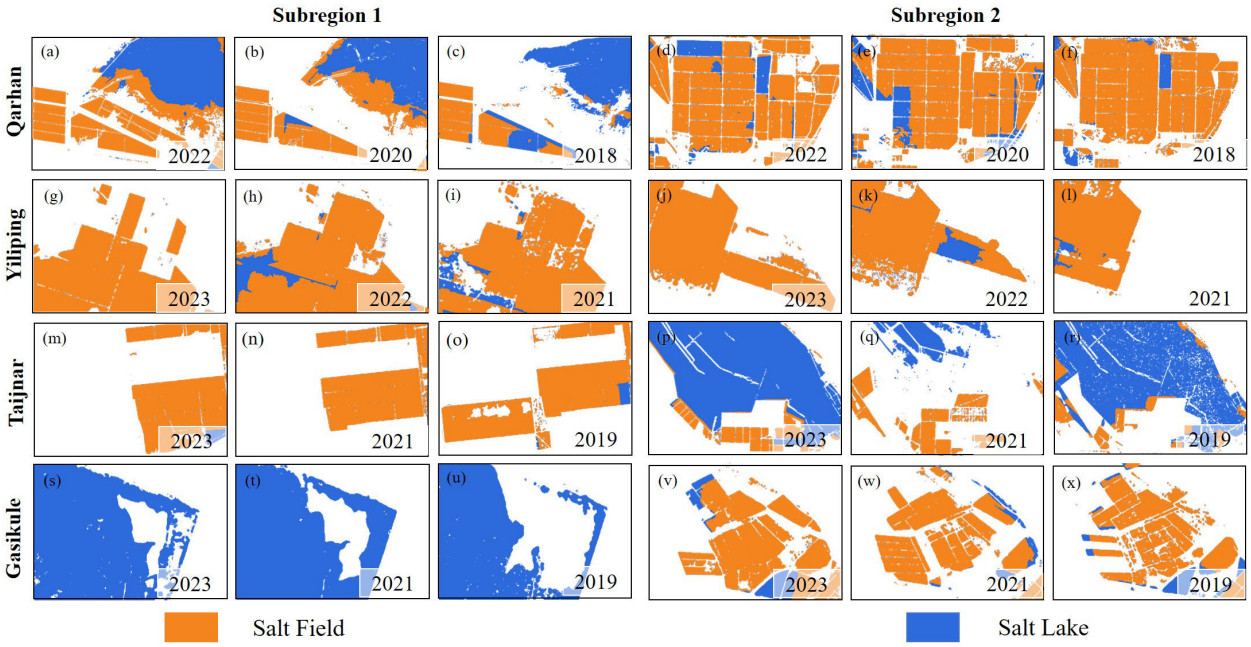


Fig. 10. Three-year mapping results for two enlarged subregions of four salt lake regions. (a)–(f) Qarhan salt lake. (g)–(i) Yiliping salt lake. (m)–(r) Taijnar salt lake. (s)–(x) Gasikule salt lake.

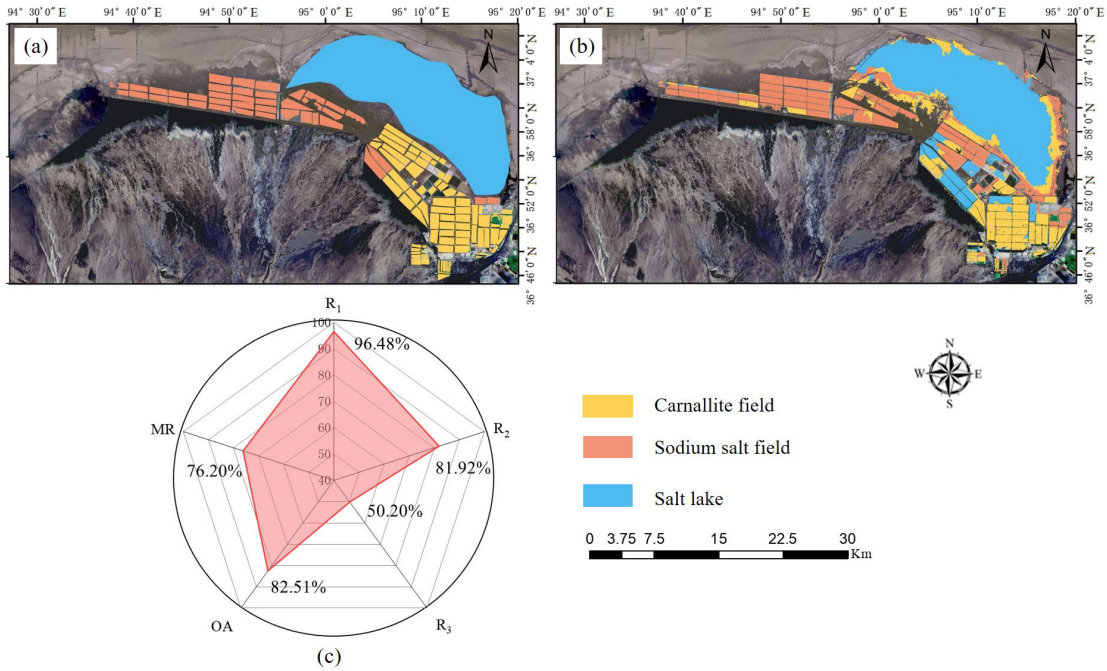


Fig. 11. Classification of Qarhan Salt Lake into Salt Lakes, Sodium Salt Fields, and Carnallite Fields with evaluation Indicators. (a) Ground truth. (b) Classification result. (c) Evaluation Indicators.

the results with the ground-truth map, it becomes evident that a considerable number of carnallite fields have been misclassified as sodium salt fields. Given that the POSD fuses multitemporal polarization features and optical features, although there is a subtle difference in multitemporal polarization features between the sodium salt fields and carnallite fields, the introduction of optical features reduces this difference. This leads to the misclassification of carnallite fields as sodium salt fields. Nonetheless, our mapping results achieve OA of 82% and MR of 76%, indicating that the mapping results are roughly consistent with the real scenario. Therefore, the proposed method has the potential to categorize the salt lake region into three categories, and the mapping results are informative.

E. Potential Use of POSD for Monitoring Salt Lake Area

Changes in the salt lake area can be detected using the proposed method. We map the changes in the area of salt lakes

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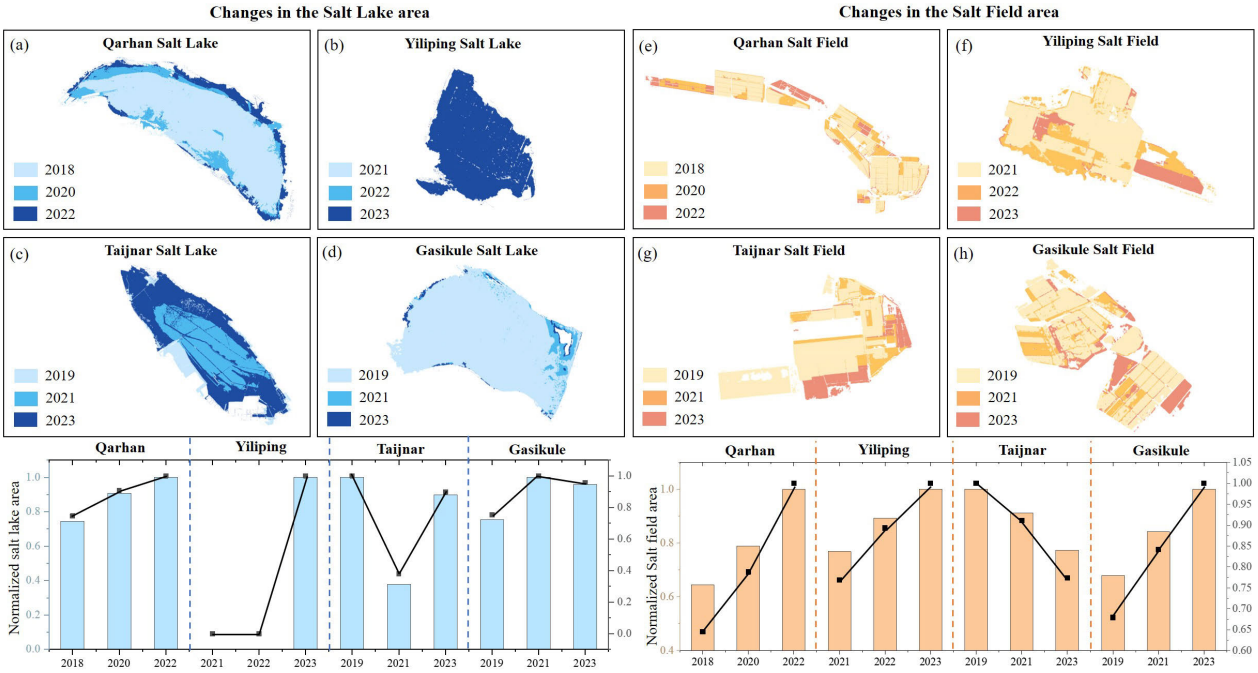


Fig. 12. Temporal variations in salt lake and salt field areas across four salt lake regions. (a) and (e) Qarhan (2018–2022). (b) and (f) Yiliping (2021–2023). (c) and (g) Taijnar (2019–2023). (d) and (h) Gasikule (2019–2023).

and salt fields over three years in each study area, as depicted in Fig. 12. Fig. 12(a) and (e) illustrates the gradual increase in the area of salt lakes and salt fields in the Qarhan salt lake. Fig. 12(b) shows the changes in the area of salt lakes in the Yiliping salt lake, clearly indicating that the salt lakes dried up in 2021 and 2022, and only reappeared in 2023. Additionally, comparing the salt fields in 2021 in the Yiliping salt lake region, there is a significant increase in the area of salt fields in 2022 and 2023, as shown in Fig. 12(f). For the Taijnar salt lake, Fig. 12(c) reveals extensive drying of the salt lakes in 2021, with a significant reduction in area compared to 2019. By 2023, there will be a notable increase in the area of the salt lakes. Fig. 12(g) illustrates the changes in the area of salt fields in the Taijnar salt lake region, showing that the distribution of salt fields varies from year to year, and the overall area is gradually decreasing. Fig. 12(d) and (h) depicts the changes in salt lakes and salt fields in the Gasikule salt lake. The area of both salt lakes and salt fields has been increasing each year, though the increase is not substantial.

F. Effect of Parameters on Experimental Results

We further explore the effect of proportion coefficients on the classification results to determine the range of optimal proportion coefficients. Specifically, we use the GA to find the proportion coefficients that will lead to the highest OA. In our case, the GA algorithm is set to iterate for 50 generations, producing 30 populations per generation. Fig. 13 depicts the proportion coefficients α and γ generated by the GA in the four study regions, and the corresponding overall classification accuracies. As can be seen in Fig. 13, for almost every pair of α and γ, a different classification result is achieved correspondingly. Besides, the overall classification accuracy

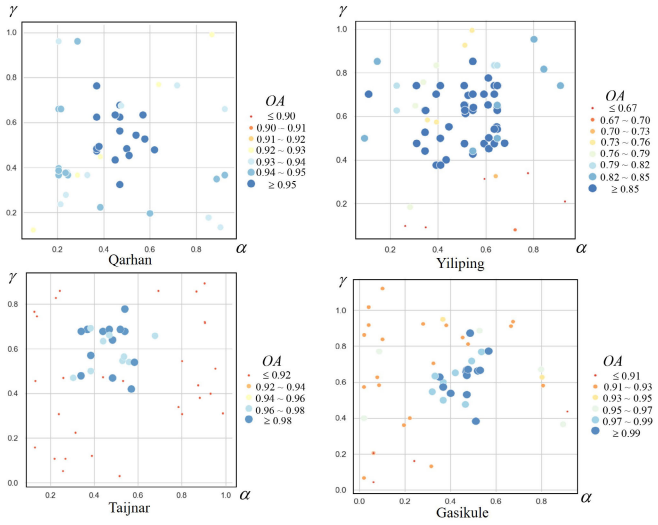


Fig. 13. Effect of different proportion coefficients on overall classification accuracy.

can be highest when α is between 0.3 and 0.6 and γ is between 0.4 and 0.7. This conclusion allows us to directly set the value of the proportion coefficients in the appropriate interval during subsequent experiments, thus avoiding the complex iterative process of the GA.

V. Conclusion

Monitoring changes in the hypersaline water body is essential for environmental protection and salt mine production. This study proposes an effective hypersaline water body classification method utilizing multitemporal PolSAR and multispectral data. Specifically, we first fuse the scattering mechanism entropy and blue bands to obtain a novel feature, termed POSD, which can capture both cumulative

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changes and ion concentrations. Then, a classifier based on the Chernoff distance of POSD is proposed, incorporating superpixels into the classification process to minimize noise effects. The proposed method is applied to classify salt lake regions across four distinct study areas. Experimental results demonstrate its superior performance compared to existing methods, including TSCA, STMA, CNNM, IOPF, MMXG, Bhat, and MinD, achieving state-of-the-art OA values of 90.91%, 92.73%, 97.40%, and 97.65% for the respective study areas. In conclusion, the proposed method effectively monitors changes in the areas of hypersaline water bodies, providing valuable guidance for environmental protection and salt mine production.

APPENDIX A

Assuming the random variable $z = \alpha x + \gamma y$. Following the definition of the cumulative distribution function (cdf), we can get

$$\begin{aligned} F(z) &= P(\alpha x + \gamma y \le z) = P\left(x \le \frac{1}{\alpha}z - \frac{\gamma}{\alpha}(y)\right) \\ &= \int_{-\infty}^{+\infty} f_{\text{msi}}(y) \int_{-\infty}^{\frac{1}{\alpha}z-\frac{\gamma}{\alpha}y} f_{\text{pol}}(x)dxdy \\ &= \int_{-\infty}^{+\infty} f_{\text{msi}}(y) F_{\text{pol}}\left(\frac{1}{\alpha}z - \frac{\gamma}{\alpha}y\right)dy \end{aligned} \tag{18}$$

where $F(\cdot)$ denotes the cumulative distribution function and $P(\cdot)$ denotes the probability. The pdf of z is obtained by taking the derivative of $F(z)$, that is,

$$f(z) = F'(z) = \frac{1}{\alpha}\int_{-\infty}^{+\infty} f_{\text{msi}}(y) f_{\text{pol}}\left(\frac{1}{\alpha}z - \frac{\gamma}{\alpha}y\right)dy. \tag{19}$$

Taking the PDFs of x and y into (19), we are able to obtain the pdf of z as

$$\begin{aligned} f(z) = F'(z) &= \frac{1}{\alpha}\int_{-\infty}^{+\infty} f_{\text{msi}}(y) f_{\text{pol}}\left(\frac{1}{\alpha}z - \frac{\gamma}{\alpha}y\right)dydy \\ &= \frac{1}{\alpha}\int_{-\infty}^{+\infty} \left\{ \begin{array}{l} \dfrac{\omega_{\text{msi1}}}{\sqrt{2\pi}\sigma_{\text{msi1}}} \exp\left[-\dfrac{(y-\mu_{\text{msi1}})^2}{2\sigma_{\text{msi1}}^2}\right] \\ +\dfrac{\omega_{\text{msi2}}}{\sqrt{2\pi}\sigma_{\text{msi2}}} \exp\left[-\dfrac{(y-\mu_{\text{msi2}})^2}{2\sigma_{\text{msi2}}^2}\right] \end{array} \right\} \\ &\quad \times \frac{1}{\sqrt{2\pi}\sigma_{\text{pol}}} \exp\left[-\frac{\left(\frac{1}{\alpha}z - \frac{\gamma}{\alpha}y - \mu_{\text{pol}}\right)^2}{2\sigma_{\text{pol}}^2}\right]dy \\ &= \frac{\omega_{\text{msi1}}}{\alpha 2\pi\sigma_{\text{msi1}}\sigma_{\text{pol}}}\int_{-\infty}^{+\infty} \\ &\quad \times \left\{ \exp -\frac{1}{2}\left[\begin{array}{l} \left(\dfrac{1}{\sigma_{\text{msi1}}^2} + \dfrac{\gamma^2}{\sigma_{\text{pol}}^2\alpha^2}\right)y^2 \\ +\left(\dfrac{2\left(\mu_{\text{pol}} - \frac{1}{\alpha}z\right)}{\sigma_{\text{pol}}^2} - \dfrac{2\mu_{\text{msi1}}}{\sigma_{\text{msi1}}^2}\right)y \\ +\dfrac{\mu_{\text{msi1}}^2}{\sigma_{\text{msi1}}^2} + \dfrac{\left(\mu_{\text{pol}} - z/\alpha\right)^2}{\sigma_{\text{pol}}^2} \end{array} \right] \right\}dy \\ &\quad + \frac{\omega_{\text{msi2}}}{\alpha 2\pi\sigma_{\text{msi2}}\sigma_{\text{pol}}}\int_{-\infty}^{+\infty} \\ &\quad \times \left\{ \exp -\frac{1}{2}\left[\begin{array}{l} \left(\dfrac{1}{\sigma_{\text{msi2}}^2} + \dfrac{\gamma^2}{\sigma_{\text{pol}}^2\alpha^2}\right)y^2 \\ +\left(\dfrac{2\left(\mu_{\text{pol}} - \frac{1}{\alpha}z\right)}{\sigma_{\text{pol}}^2} - \dfrac{2\mu_{\text{msi2}}}{\sigma_{\text{msi2}}^2}\right)y \\ +\dfrac{\mu_{\text{msi2}}^2}{\sigma_{\text{msi2}}^2} + \dfrac{\left(\mu_{\text{pol}} - z/\alpha\right)^2}{\sigma_{\text{pol}}^2} \end{array} \right] \right\}dy \\ &= \frac{\omega_{\text{msi1}}}{\sqrt{2\pi}\sqrt{\sigma_{\text{pol}}^2\alpha^2 + \gamma^2\sigma_{\text{msi1}}^2}} \\ &\quad \times \exp\left[-\frac{\left(z - \alpha\mu_{\text{pol}} - \gamma\mu_{\text{msi1}}\right)^2}{2\left(\sigma_{\text{pol}}^2\alpha^2 + \gamma^2\sigma_{\text{msi1}}^2\right)}\right] \\ &\quad \times \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi}\sqrt{\frac{\sigma_{\text{msi1}}^2\sigma_{\text{pol}}^2\alpha^2}{\alpha^2\sigma_{\text{pol}}^2+\gamma^2\sigma_{\text{msi1}}^2}}} \\ &\quad \times \exp\left\{ -\frac{\alpha^2\sigma_{\text{pol}}^2 + \gamma^2\sigma_{\text{msi1}}^2}{2\sigma_{\text{msi1}}^2\sigma_{\text{pol}}^2\alpha^2} \right. \\ &\qquad \left. \times \left[y - \frac{\mu_{\text{msi1}}\sigma_{\text{pol}}^2\alpha^2 + \gamma\left(z - \alpha\mu_{\text{pol}}\right)\sigma_{\text{msi1}}^2}{\gamma^2\left(\sigma_{\text{pol}}^2 + \sigma_{\text{msi1}}^2\right)}\right] \right\}dy \\ &\quad + \frac{\omega_{\text{msi2}}}{\sqrt{2\pi}\sqrt{\sigma_{\text{pol}}^2\alpha^2 + \gamma^2\sigma_{\text{msi2}}^2}} \\ &\quad \times \exp\left[-\frac{\left(z - \alpha\mu_{\text{pol}} - \gamma\mu_{\text{msi2}}\right)^2}{2\left(\sigma_{\text{pol}}^2\alpha^2 + \gamma^2\sigma_{\text{msi2}}^2\right)}\right] \\ &\quad \times \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi}\sqrt{\frac{\sigma_{\text{msi2}}^2\sigma_{\text{pol}}^2\alpha^2}{\alpha^2\sigma_{\text{pol}}^2+\gamma^2\sigma_{\text{msi2}}^2}}} \\ &\quad \times \exp\left\{ -\frac{\alpha^2\sigma_{\text{pol}}^2 + \gamma^2\sigma_{\text{msi2}}^2}{2\sigma_{\text{msi2}}^2\sigma_{\text{pol}}^2\alpha^2} \right. \\ &\qquad \left. \times \left[y - \frac{\mu_{\text{msi2}}\sigma_{\text{pol}}^2\alpha^2 + \gamma\left(z - \alpha\mu_{\text{pol}}\right)\sigma_{\text{msi2}}^2}{\gamma^2\left(\sigma_{\text{pol}}^2 + \sigma_{\text{msi2}}^2\right)}\right] \right\}dy \\ &= \frac{\omega_{\text{msi1}}}{\sqrt{2\pi}\sqrt{\sigma_{\text{pol}}^2\alpha^2 + \gamma^2\sigma_{\text{msi1}}^2}} \\ &\quad \times \exp\left[-\frac{\left(z - \alpha\mu_{\text{pol}} - \gamma\mu_{\text{msi1}}\right)^2}{2\left(\sigma_{\text{pol}}^2\alpha^2 + \gamma^2\sigma_{\text{msi1}}^2\right)}\right] \\ &\quad + \frac{\omega_{\text{msi2}}}{\sqrt{2\pi}\sqrt{\sigma_{\text{pol}}^2\alpha^2 + \gamma^2\sigma_{\text{msi2}}^2}} \\ &\quad \times \exp\left[-\frac{\left(z - \alpha\mu_{\text{pol}} - \gamma\mu_{\text{msi2}}\right)^2}{2\left(\sigma_{\text{pol}}^2\alpha^2 + \gamma^2\sigma_{\text{msi2}}^2\right)}\right]. \end{aligned} \tag{20}$$

APPENDIX B

To calculate the Chernoff distance under a Gaussian mixture distribution, we first take the two PDFs into the integral

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in (13)

$$\begin{aligned}&\int f_1^{\beta}(z) f_2^{1-\beta}(z)dz\\&=\int\left\{\frac{\tilde{\omega}_1}{\sqrt{2\pi}\tilde{\sigma}_1}\exp\left[-\frac{(z-\tilde{\mu}_1)^2}{2\tilde{\sigma}_1^2}\right]\right.\\&\qquad\left.+\frac{\tilde{\omega}_2}{\sqrt{2\pi}\tilde{\sigma}_2}\exp\left[-\frac{(z-\tilde{\mu}_2)^2}{2\tilde{\sigma}_2^2}\right]\right\}^{\beta}\\&\quad\times\left\{\frac{\bar{\omega}_1}{\sqrt{2\pi}\bar{\sigma}_1}\exp\left[-\frac{(z-\bar{\mu}_1)^2}{2\bar{\sigma}_1^2}\right]\right.\\&\qquad\left.+\frac{\bar{\omega}_2}{\sqrt{2\pi}\bar{\sigma}_2}\exp\left[-\frac{(z-\bar{\mu}_2)^2}{2\bar{\sigma}_2^2}\right]\right\}^{1-\beta}dz.\end{aligned}\tag{21}$$

The powers of the Gaussian mixture distribution can be computed by a first-order expansion approximation [38], that is,

$$\begin{aligned}f_1^{\beta}(z)&=\left\{\frac{\tilde{\omega}_1}{\sqrt{2\pi}\tilde{\sigma}_1}\exp\left[-\frac{(z-\tilde{\mu}_1)^2}{2\tilde{\sigma}_1^2}\right]\right.\\&\qquad\left.+\frac{\tilde{\omega}_2}{\sqrt{2\pi}\tilde{\sigma}_2}\exp\left[-\frac{(z-\tilde{\mu}_2)^2}{2\tilde{\sigma}_2^2}\right]\right\}^{\beta}\\&\approx\frac{1}{\tilde{\omega}_1+\tilde{\omega}_2}\left\{\frac{\tilde{\omega}_1^{\beta}}{\sqrt{2\pi}\frac{\tilde{\sigma}_1}{\sqrt{\beta}}}\exp\left[-\frac{(z-\tilde{\mu}_1)^2}{2\frac{\tilde{\sigma}_1^2}{\beta}}\right]\right.\\&\qquad\left.+\frac{\tilde{\omega}_2^{\beta}}{\sqrt{2\pi}\frac{\tilde{\sigma}_2}{\sqrt{\beta}}}\exp\left[-\frac{(z-\tilde{\mu}_2)^2}{2\frac{\tilde{\sigma}_2^2}{\beta}}\right]\right\}\\&=\frac{1}{\tilde{\omega}_1+\tilde{\omega}_2}\left\{\frac{\tilde{\omega}_1^{\beta}}{\sqrt{2\pi}\tilde{\Sigma}_1}\exp\left[-\frac{(z-\tilde{\mu}_1)^2}{2\tilde{\Sigma}_1^2}\right]\right.\\&\qquad\left.+\frac{\tilde{\omega}_2^{\beta}}{\sqrt{2\pi}\tilde{\Sigma}_2}\exp\left[-\frac{(z-\tilde{\mu}_2)^2}{2\tilde{\Sigma}_2^2}\right]\right\}\end{aligned}\tag{22}$$

$$\begin{aligned}f_2^{1-\beta}(z)&=\left\{\frac{\bar{\omega}_1}{\sqrt{2\pi}\bar{\sigma}_1}\exp\left[-\frac{(z-\bar{\mu}_1)^2}{2\bar{\sigma}_1^2}\right]\right.\\&\qquad\left.+\frac{\bar{\omega}_2}{\sqrt{2\pi}\bar{\sigma}_2}\exp\left[-\frac{(z-\bar{\mu}_2)^2}{2\bar{\sigma}_2^2}\right]\right\}^{1-\beta}\\&\approx\frac{1}{\bar{\omega}_1+\bar{\omega}_2}\left\{\frac{\bar{\omega}_1^{(1-\beta)}}{\sqrt{2\pi}\frac{\bar{\sigma}_1}{\sqrt{1-\beta}}}\exp\left[-\frac{(z-\bar{\mu}_1)^2}{2\frac{\bar{\sigma}_1^2}{1-\beta}}\right]\right.\\&\qquad\left.+\frac{\bar{\omega}_2^{(1-\beta)}}{\sqrt{2\pi}\frac{\bar{\sigma}_2}{\sqrt{1-\beta}}}\exp\left[-\frac{(z-\bar{\mu}_2)^2}{2\frac{\bar{\sigma}_2^2}{1-\beta}}\right]\right\}\\&=\frac{1}{\bar{\omega}_1+\bar{\omega}_2}\left\{\frac{\bar{\omega}_1^{(1-\beta)}}{\sqrt{2\pi}\bar{\Sigma}_1}\exp\left[-\frac{(z-\bar{\mu}_1)^2}{2\bar{\Sigma}_1^2}\right]\right.\\&\qquad\left.+\frac{\bar{\omega}_2^{(1-\beta)}}{\sqrt{2\pi}\bar{\Sigma}_2}\exp\left[-\frac{(z-\bar{\mu}_2)^2}{2\bar{\Sigma}_2^2}\right]\right\}\end{aligned}\tag{23}$$

where $\tilde{\Sigma}_1=\tilde{\sigma}_1/\sqrt{\beta}$, $\tilde{\Sigma}_2=\tilde{\sigma}_2/\sqrt{\beta}$, $\bar{\Sigma}_1=\bar{\sigma}_1/(1-\beta)^{1/2}$ and $\bar{\Sigma}_2=\bar{\sigma}_2/(1-\beta)^{1/2}$. Substituting (22) and (23) into (21), we can obtain

$$\begin{aligned}&\int f_1^{\beta}(z) f_2^{1-\beta}(z)dz\\&=\frac{1}{(\tilde{\omega}_1+\tilde{\omega}_2)(\bar{\omega}_1+\bar{\omega}_2)}\\&\quad\times\int\left\{\frac{\tilde{\omega}_1^{\beta}}{\sqrt{2\pi}\tilde{\Sigma}_1}\exp\left[-\frac{(z-\tilde{\mu}_1)^2}{2\tilde{\Sigma}_1^2}\right]\right.\\&\qquad\left.+\frac{\tilde{\omega}_2^{\beta}}{\sqrt{2\pi}\tilde{\Sigma}_2}\exp\left[-\frac{(z-\tilde{\mu}_2)^2}{2\tilde{\Sigma}_2^2}\right]\right\}\\&\quad\times\left\{\frac{\bar{\omega}_1^{(1-\beta)}}{\sqrt{2\pi}\bar{\Sigma}_1}\exp\left[-\frac{(z-\bar{\mu}_1)^2}{2\bar{\Sigma}_1^2}\right]\right.\\&\qquad\left.+\frac{\bar{\omega}_2^{(1-\beta)}}{\sqrt{2\pi}\bar{\Sigma}_2}\exp\left[-\frac{(z-\bar{\mu}_2)^2}{2\bar{\Sigma}_2^2}\right]\right\}dz\\&=\frac{1}{(\tilde{\omega}_1+\tilde{\omega}_2)(\bar{\omega}_1+\bar{\omega}_2)}\\&\quad\times\left\{\int\frac{\tilde{\omega}_1^{\beta}}{\sqrt{2\pi}\tilde{\Sigma}_1}\exp\left[-\frac{(z-\tilde{\mu}_1)^2}{2\tilde{\Sigma}_1^2}\right]\right.\\&\qquad\times\frac{\bar{\omega}_1^{(1-\beta)}}{\sqrt{2\pi}\bar{\Sigma}_1}\exp\left[-\frac{(z-\bar{\mu}_1)^2}{2\bar{\Sigma}_1^2}\right]dz+\int\frac{\tilde{\omega}_1^{\beta}}{\sqrt{2\pi}\tilde{\Sigma}_1}\\&\qquad\times\exp\left[-\frac{(z-\tilde{\mu}_1)^2}{2\tilde{\Sigma}_1^2}\right]\times\frac{\bar{\omega}_2^{(1-\beta)}}{\sqrt{2\pi}\bar{\Sigma}_2}\\&\qquad\times\exp\left[-\frac{(z-\bar{\mu}_2)^2}{2\bar{\Sigma}_2^2}\right]dz\\&\qquad+\int\frac{\tilde{\omega}_2^{\beta}}{\sqrt{2\pi}\tilde{\Sigma}_2}\exp\left[-\frac{(z-\tilde{\mu}_2)^2}{2\tilde{\Sigma}_2^2}\right]\\&\qquad\times\frac{\bar{\omega}_1^{(1-\beta)}}{\sqrt{2\pi}\bar{\Sigma}_1}\exp\left[-\frac{(z-\bar{\mu}_1)^2}{2\bar{\Sigma}_1^2}\right]dz\\&\qquad+\int\frac{\tilde{\omega}_2^{\beta}}{\sqrt{2\pi}\tilde{\Sigma}_2}\exp\left[-\frac{(z-\tilde{\mu}_2)^2}{2\tilde{\Sigma}_2^2}\right]\\&\qquad\left.\times\frac{\bar{\omega}_2^{(1-\beta)}}{\sqrt{2\pi}\bar{\Sigma}_2}\exp\left[-\frac{(z-\bar{\mu}_2)^2}{2\bar{\Sigma}_2^2}\right]dz\right\}.\end{aligned}\tag{24}$$

The four integrals in (24) can be calculated using the same method. Taking the first integral as an example, the detailed calculation procedure is as follows:

$$\begin{aligned}&\int\frac{\tilde{\omega}_1^{\beta}}{\sqrt{2\pi}\tilde{\Sigma}_1}\exp\left[-\frac{(z-\tilde{\mu}_1)^2}{2\tilde{\Sigma}_1^2}\right]\times\frac{\bar{\omega}_1^{(1-\beta)}}{\sqrt{2\pi}\bar{\Sigma}_1}\\&\quad\times\exp\left[-\frac{(z-\bar{\mu}_1)^2}{2\bar{\Sigma}_1^2}\right]dz\\&=\frac{\tilde{\omega}_1^{\beta}\bar{\omega}_1^{(1-\beta)}}{2\pi\tilde{\Sigma}_1\bar{\Sigma}_1}\\&\quad\times\int\exp\left[-\left((z-\tilde{\mu}_1)^2\bar{\Sigma}_1^2+(z-\bar{\mu}_1)^2\tilde{\Sigma}_1^2\right)/2\tilde{\Sigma}_1^2\bar{\Sigma}_1^2\right]dz\\&=\frac{\tilde{\omega}_1^{\beta}\bar{\omega}_1^{(1-\beta)}}{2\pi\tilde{\Sigma}_1\bar{\Sigma}_1}\int\exp\left[-\left(z^2-2\frac{\tilde{\mu}_1\bar{\Sigma}_1^2+\bar{\mu}_1\tilde{\Sigma}_1^2}{\tilde{\Sigma}_1^2+\bar{\Sigma}_1^2}z\right.\right.\end{aligned}$$

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$$+\frac{\tilde{\mu}_1^2\bar{\Sigma}_1^2+\bar{\mu}_1^2\tilde{\Sigma}_1^2}{\tilde{\Sigma}_1^2+\bar{\Sigma}_1^2}\Bigg)\Bigg/\frac{2\tilde{\Sigma}_1^2\bar{\Sigma}_1^2}{\tilde{\Sigma}_1^2+\bar{\Sigma}_1^2}\Bigg]dz$$
$$=\frac{\tilde{\omega}_1^{\beta}\bar{\omega}_1^{(1-\beta)}}{2\pi\tilde{\Sigma}_1\bar{\Sigma}_1}\int\exp\Bigg[-\Bigg(z-\frac{\tilde{\mu}_1\bar{\Sigma}_1^2+\bar{\mu}_1\tilde{\Sigma}_1^2}{\tilde{\Sigma}_1^2+\bar{\Sigma}_1^2}\Bigg)^2\Bigg/\frac{2\tilde{\Sigma}_1^2\bar{\Sigma}_1^2}{\tilde{\Sigma}_1^2+\bar{\Sigma}_1^2}$$
$$-\frac{\left(\tilde{\mu}_1-\bar{\mu}_1\right)^2}{2\left(\tilde{\Sigma}_1^2+\bar{\Sigma}_1^2\right)}\Bigg]dz$$
$$=\frac{\tilde{\omega}_1^{\beta}\bar{\omega}_1^{(1-\beta)}}{\sqrt{2\pi\left(\tilde{\Sigma}_1^2+\bar{\Sigma}_1^2\right)}}\exp\left[-\frac{\left(\tilde{\mu}_1-\bar{\mu}_1\right)^2}{2\left(\tilde{\Sigma}_1^2+\bar{\Sigma}_1^2\right)}\right]\frac{1}{\sqrt{2\pi}\frac{\tilde{\Sigma}_1\bar{\Sigma}_1}{\sqrt{\tilde{\Sigma}_1^2+\bar{\Sigma}_1^2}}}$$
$$\times\int\exp\left[-\left(z-\frac{\tilde{\mu}_1\bar{\Sigma}_1^2+\bar{\mu}_1\tilde{\Sigma}_1^2}{\tilde{\Sigma}_1^2+\bar{\Sigma}_1^2}\right)^2/\frac{2\tilde{\Sigma}_1^2\bar{\Sigma}_1^2}{\tilde{\Sigma}_1^2+\bar{\Sigma}_1^2}\right]dz$$
$$=\frac{\tilde{\omega}_1^{\beta}\bar{\omega}_1^{(1-\beta)}}{\sqrt{2\pi\left(\tilde{\Sigma}_1^2+\bar{\Sigma}_1^2\right)}}\exp\left[-\frac{\left(\tilde{\mu}_1-\bar{\mu}_1\right)^2}{2\left(\tilde{\Sigma}_1^2+\bar{\Sigma}_1^2\right)}\right]. \tag{25}$$

Similarly, we can compute the other integrals in (24). Bringing these integrals and $\tilde{\Sigma}_1=\tilde{\sigma}_1/\sqrt{\beta}$, $\tilde{\Sigma}_2=\tilde{\sigma}_2/\sqrt{\beta}$, $\bar{\Sigma}_1=\bar{\sigma}_1/(1-\beta)^{1/2}$, $\bar{\Sigma}_2=\bar{\sigma}_2/(1-\beta)^{1/2}$ into (24), and then, taking the results into (10), we can get we can obtain the Chernoff distance of the two Gaussian mixture distributions as follows:

$$\begin{aligned}
&D(f_1,f_2)\\
&=\ln\left(\tilde{\omega}_1+\tilde{\omega}_2\right)\left(\bar{\omega}_1+\bar{\omega}_2\right)\\
&\quad-\ln\Bigg\{\frac{\tilde{\omega}_1^{\beta}\bar{\omega}_1^{(1-\beta)}}{\sqrt{2\pi\left(\tilde{\sigma}_1^2/\beta+\bar{\sigma}_1^2/1-\beta\right)}}\\
&\qquad\times\exp\left[-\frac{\left(\tilde{\mu}_1-\bar{\mu}_1\right)^2}{2\left(\tilde{\sigma}_1^2/\beta+\bar{\sigma}_1^2/1-\beta\right)}\right]\\
&\qquad+\frac{\tilde{\omega}_1^{\beta}\bar{\omega}_2^{(1-\beta)}}{\sqrt{2\pi\left(\tilde{\sigma}_1^2/\beta+\bar{\sigma}_2^2/1-\beta\right)}}\\
&\qquad\times\exp\left[-\frac{\left(\tilde{\mu}_1-\bar{\mu}_2\right)^2}{2\left(\tilde{\sigma}_1^2/\beta+\bar{\sigma}_2^2/1-\beta\right)}\right]\\
&\qquad+\frac{\tilde{\omega}_2^{\beta}\bar{\omega}_1^{(1-\beta)}}{\sqrt{2\pi\left(\tilde{\sigma}_2^2/\beta+\bar{\sigma}_1^2/1-\beta\right)}}\\
&\qquad\times\exp\left[-\frac{\left(\tilde{\mu}_2-\bar{\mu}_1\right)^2}{2\left(\tilde{\sigma}_2^2/\beta+\bar{\sigma}_1^2/1-\beta\right)}\right]\\
&\qquad+\frac{\tilde{\omega}_2^{\beta}\bar{\omega}_2^{(1-\beta)}}{\sqrt{2\pi\left(\tilde{\sigma}_2^2/\beta+\bar{\sigma}_2^2/1-\beta\right)}}\\
&\qquad\times\exp\left[-\frac{\left(\tilde{\mu}_2-\bar{\mu}_2\right)^2}{2\left(\tilde{\sigma}_2^2/\beta+\bar{\sigma}_2^2/1-\beta\right)}\right]\Bigg\}.
\end{aligned}\tag{26}$$

REFERENCES

[1] A. Crawford et al., "Lithium: The big picture," *One Earth*, vol. 4, no. 3, pp. 323–326, Mar. 2021, doi: 10.1016/j.oneear.2021.02.021.

[2] D. V. Chapman and T. Sullivan, "The role of water quality monitoring in the sustainable use of ambient waters," *One Earth*, vol. 5, no. 2, pp. 132–137, Feb. 2022, doi: 10.1016/j.oneear.2022.01.008.

[3] R. Kong, F. Xue, J. Wang, H. Zhai, and L. Zhao, "Research on mineral resources and environment of salt lakes in Qinghai province based on system dynamics theory," *Resour. Policy*, vol. 52, pp. 19–28, Jun. 2017, doi: 10.1016/j.resourpol.2017.01.006.

[4] F. Ma, F. Zhang, D. Xiang, Q. Yin, and Y. Zhou, "Fast task-specific region merging for SAR image segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5222316, doi: 10.1109/TGRS.2022.3141125.

[5] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3431–3440.

[6] H. Zhao, J. Shi, X. Qi, X. Wang, and J. Jia, "Pyramid scene parsing network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2881–2890.

[7] F. Zhang, X. Sun, F. Ma, and Q. Yin, "Superpixelwise likelihood ratio test statistic for PolSAR data and its application to built-up area extraction," *ISPRS J. Photogramm. Remote Sens.*, vol. 209, pp. 233–248, Mar. 2024, doi: 10.1016/j.isprsjprs.2024.02.009.

[8] M. Dabboor and M. Shokr, "A new likelihood ratio for supervised classification of fully polarimetric SAR data: An application for sea ice type mapping," *ISPRS J. Photogramm. Remote Sens.*, vol. 84, pp. 1–11, Oct. 2013.

[9] Q. Yin et al., "Coherence matrix power model for scattering variation representation in multi-temporal PolSAR crop classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 9797–9810, 2024, doi: 10.1109/JSTARS.2024.3395689.

[10] J. Ni, C. López-Martínez, Z. Hu, and F. Zhang, "Multitemporal SAR and polarimetric SAR optimization and classification: Reinterpreting temporal coherence," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5236617, doi: 10.1109/TGRS.2022.3214097.

[11] L. Gao, Z. Lin, Q. Yin, and W. Hong, "Classification performance comparison of time-variant scattering features of multi-temporal polarimetric SAR data," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2023, pp. 8050–8053, doi: 10.1109/IGARSS52108.2023.10282315.

[12] F. Zhang, F. Meng, F. Ma, Q. Yin, and Y. Zhou, "Time correlation entropy: A novel multitemporal PolSAR feature and its application in salt lake classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4506814, doi: 10.1109/TGRS.2024.3404598.

[13] S. S. Kaushal et al., "The anthropogenic salt cycle," *Nature Rev. Earth Environ.*, vol. 4, no. 11, pp. 770–784, Oct. 2023, doi: 10.1038/s43017-023-00485-y.

[14] N. G. Kasapoglu and T. Eltoft, "Decision fusion of classifiers for multifrequency PolSAR and optical data classification," in *Proc. 6th Int. Conf. Recent Adv. Space Technol. (RAST)*, Istanbul, Turkey, Jun. 2013, pp. 411–416, doi: 10.1109/RAST.2013.6581242.

[15] N. Maleki, Y. Zeinali, and S. T. A. Niaki, "A k-NN method for lung cancer prognosis with the use of a genetic algorithm for feature selection," *Expert Syst. Appl.*, vol. 164, Feb. 2021, Art. no. 113981, doi: 10.1016/j.eswa.2020.113981.

[16] H. Zhang, "Research of the target detection based on the statistical distribution characteristics of multispectral filtering," College Inf. Commun. Eng., Harbin Eng. Univ., Harbin, China, Tech. Rep., 2012.

[17] V. M. Panaretos and Y. Zemel, "Statistical aspects of Wasserstein distances," in *Proc. 19th IEEE Int. Conf. Image Process.*, Orlando, FL, USA, Aug. 2012, pp. 2093–2096, doi: 10.1109/ICIP.2012.6467304.

[18] M. Dabboor, M. J. Collins, V. Karathanassi, and A. Braun, "An unsupervised classification approach for polarimetric SAR data based on the Chernoff distance for the complex Wishart distribution," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, pp. 4200–4213, 2013.

[19] R. Duda, P. Hart, and D. Stork, *Pattern Classification*. New York, NY, USA: Wiley, 2000.

[20] F. Ma, D. Xiang, K. Yang, Q. Yin, and F. Zhang, "Weakly supervised deep soft clustering for flood identification in SAR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022, doi: 10.1109/LGRS.2022.3150778.

[21] Y. He, S. Wang, and X. Gao, "Analysis and research of spinal CT image segmentation based on improved watershed algorithm," in *Proc. IEEE 3rd Int. Conf. Electron. Commun., Internet Things Big Data (ICEIB)*, Taichung, Taiwan, Apr. 2023, pp. 119–123, doi: 10.1109/ICEIB57887.2023.10170668.

[22] B. Panić, J. Klemenc, and M. Nagode, "Improved initialization of the EM algorithm for mixture model parameter estimation," *Mathematics*, vol. 8, no. 3, p. 373, Mar. 2020, doi: 10.3390/math8030373.

[23] X. Pi et al., "Mapping global lake dynamics reveals the emerging roles of small lakes," *Nature Commun.*, vol. 13, no. 1, pp. 1–12, Oct. 2022, doi: 10.1038/s41467-022-33239-3.

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[24] R. Li et al., "Simulation study on the mining conditions of dissolution of low grade solid potash ore in qarhan salt lake," *Sci. Rep.*, vol. 11, no. 1, pp. 1–9, May 2021, doi: 10.1038/s41598-021-88818-z.
[25] C. Liang et al., "Salinization mechanism of lakes and controls on organic matter enrichment: From present to deep-time records," *Earth-Sci. Rev.*, vol. 251, Apr. 2024, Art. no. 104720, doi: 10.1016/j.earscirev.2024.104720.
[26] Y. Zhang, Y. Hu, L. Wang, and W. Sun, "Systematic review of lithium extraction from salt-lake brines via precipitation approaches," *Minerals Eng.*, vol. 139, Aug. 2019, Art. no. 105868, doi: 10.1016/j.mineng.2019.105868.
[27] J. Wang and Y. Zhao, "High resolution remote sensing information identification for the salinity in GASIKULE salt lake," *Minerals Eng.*, vol. 21, pp. 1–6, Jun. 2012, doi: 10.1109/rsete.2012.6260585.
[28] F. Ma, X. Sun, F. Zhang, Y. Zhou, and H.-C. Li, "What catch your attention in SAR images: Saliency detection based on soft-superpixel lacunarity cue," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5200817, doi: 10.1109/TGRS.2022.3231253.
[29] N. Yagüe-Martínez et al., "Interferometric processing of Sentinel-1 Tops data," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 4, pp. 2220–2234, Apr. 2016, doi: 10.1109/TGRS.2015.2497902.
[30] J. S. Czapla-Myers, K. J. Thome, N. J. Anderson, L. Leigh, C. T. Pinto, and B. N. Wenny, "The ground-based absolute radiometric calibration of the Landsat 9 operational land imager," *Remote Sens.*, vol. 16, no. 6, pp. 2072–4292, Mar. 2024. [Online]. Available: https://www.mdpi.com/2072-4292/16/6/1101
[31] Y. Guo, N. Xu, and Z. Jin, "Structured kernel interpolation for scalable mixture of Gaussian processes," in *Proc. 7th Int. Conf. Intell. Comput. Signal Process. (ICSP)*, Apr. 2022, pp. 61–65, doi: 10.1109/ICSP54964.2022.9778760.
[32] A. Alonso-González, C. López-Martínez, K. P. Papathanassiou, and I. Hajnsek, "Polarimetric SAR time series change analysis over agricultural areas," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 10, pp. 7317–7330, Oct. 2020.
[33] Z. Han, C. Zhang, L. Gao, Z. Zeng, B. Zhang, and P. M. Atkinson, "Spatio-temporal multi-level attention crop mapping method using time-series SAR imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 206, pp. 293–310, Dec. 2023, doi: 10.1016/j.isprsjprs.2023.11.016.
[34] S. Preethi, A. A. Prakash, R. Ramyea, K. Rahul, K. Ramnath, and C. J. Pravenkumar, "CNN based automated land use classification from remotely sensed image," in *Proc. Innov. Power Adv. Comput. Technol. (i-PACT)*, Nov. 2021, pp. 1–4, doi: 10.1109/i-pact52855.2021.9696772.
[35] Y. Zhao and M. Jiang, "Integration of optical and SAR imagery for dual PolSAR features optimization and land cover mapping," *IEEE J. Miniaturization Air Space Syst.*, vol. 3, no. 2, pp. 67–76, Jun. 2022, doi: 10.1109/JMASS.2022.3195955.
[36] B. A. S. Pratama, P. Danoedoro, and S. Arjasakusuma, "Exploring optimal integration schemes for Sentinel-1 SAR and Sentinel-2 multispectral data in land cover mapping across different atmospheric conditions," *Remote Sens. Appl., Soc. Environ.*, vol. 34, Mar. 2024, Art. no. 101185, doi: 10.1016/j.rsase.2024.101185.
[37] J. Tang, J. Cheng, D. Xiang, and C. Hu, "Large-difference-scale target detection using a revised Bhattacharyya distance in SAR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, 2022, Art. no. 4506205, doi: 10.1109/LGRS.2022.3161931.
[38] K. Lu, R. Zhou, and J. Zhang, "Approximate Chernoff fusion of Gaussian mixtures for ballistic target tracking in the re-entry phase," *Aerosp. Sci. Technol.*, vol. 61, pp. 21–28, Feb. 2017, doi: 10.1016/j.ast.2016.11.015.



Fan Zhang (Senior Member, IEEE) received the B.E. degree in communication engineering from the Civil Aviation University of China, Tianjin, China, in 2002, the M.S. degree in signal and information processing from Beihang University, Beijing, China, in 2005, and the Ph.D. degree in signal and information processing from the Institute of Electronics, Chinese Academy of Sciences, Beijing, in 2008.

He is currently a Full Professor in electronic and information engineering with Beijing University of Chemical Technology, Beijing. His research interests are remote sensing image processing, high-performance computing, and artificial intelligence.

Dr. Zhang is an Associate Editor of IEEE ACCESS and a reviewer for IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING, IEEE GEOSCIENCE AND REMOTE SENSING LETTERS, and *Journal of Radars*.



Fanle Meng is currently pursuing the master's degree with Beijing University of Chemical Technology, Beijing, China. He is majoring in electronic information technology.

His research interests include synthetic aperture radar (SAR) image processing and classification of land covers.



Fei Ma (Member, IEEE) received the B.S., M.S., and Ph.D. degrees in electronic and information engineering from Beihang University, Beijing, China, in 2013, 2016, and 2020, respectively.

From 2017 to 2018, he was a Research Fellow with the Department of Electrical Engineering, McGill University, Montreal, QC, Canada. He is currently a Full Associate Professor at the College of Information Science and Technology, Beijing University of Chemical Technology, Beijing. His research interests include synthetic aperture radar (SAR) image processing, machine learning, artificial intelligence, and target detection.



Qiang Yin (Senior Member, IEEE) received the M.Eng. and Ph.D. degrees in signal and information processing from the Institute of Electronics, Chinese Academy of Science, Beijing, China, in 2008 and 2016, respectively.

In 2014, she visited the European Space Agency, Rome, Italy, for one year. She is currently a Professor of electronic and information engineering at Beijing University of Chemical Technology, Beijing. Her general research interests lie in radar remote sensing image processing, machine learning classification of polarized features, applications of geosynthesis modeling, and quantitative inversion.



Yongsheng Zhou (Member, IEEE) received the B.E. degree in communication engineering from Beijing Information Science and Technology University, Beijing, China, in 2005, and the Ph.D. degree in signal and information processing from the Institute of Electronics, Chinese Academy of Sciences, Beijing, in 2010.

He is currently a Professor of electronic and information engineering at Beijing University of Chemical Technology, Beijing. His general research interests lie in target detection and recognition from microwave remotely sensed images, digital signals, and image processing.

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